The Visualization shows that the positive trend between unitprice and exact sale amount features with respect to quantity. It means when quantity increases the exact sale amount also increases.

Amazon video prime DataFrame

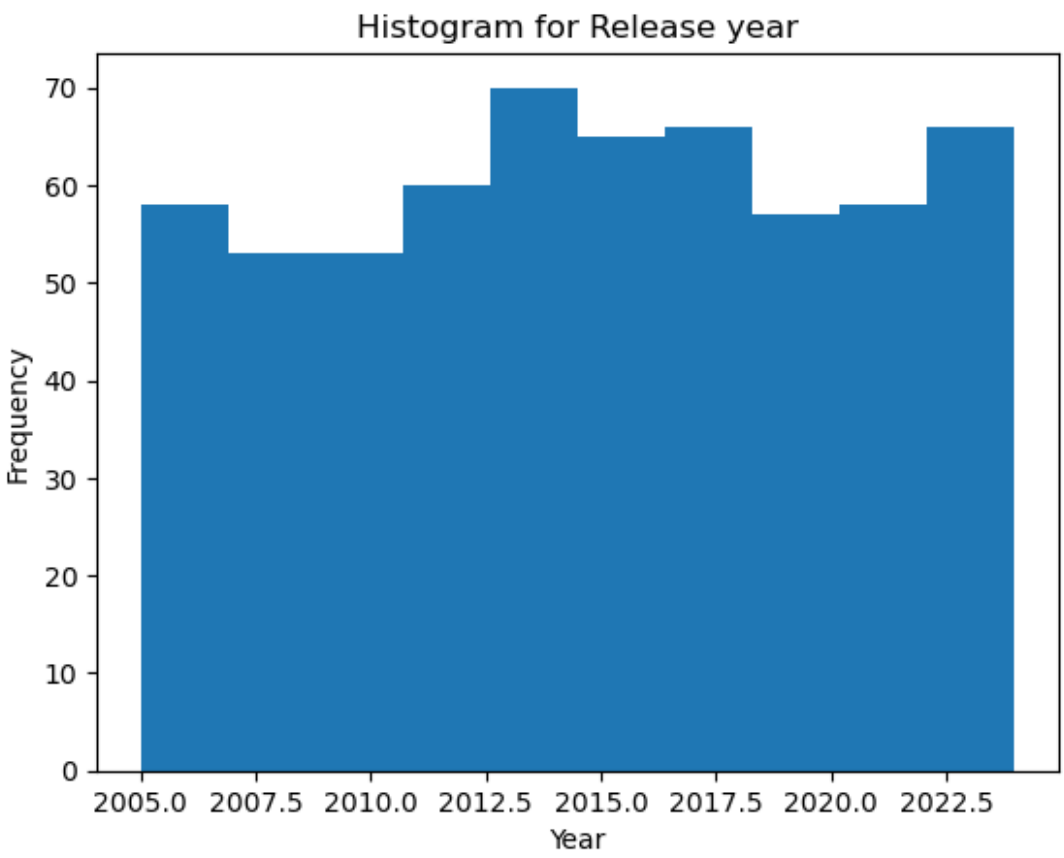
```
df=pd.read_csv("Downloads/amazon_prime_video_data.csv")

df.dtypes
```

```
content_id              int64
content_type           object
genre                  object
language               object
release_year            int64
duration_minutes        int64
imdb_rating           float64
age_rating             object
country                object
added_date             object
views_millions        float64
avg_watch_time_min    float64
likes_millions        float64
subscription_plan      object
is_trending            object
```

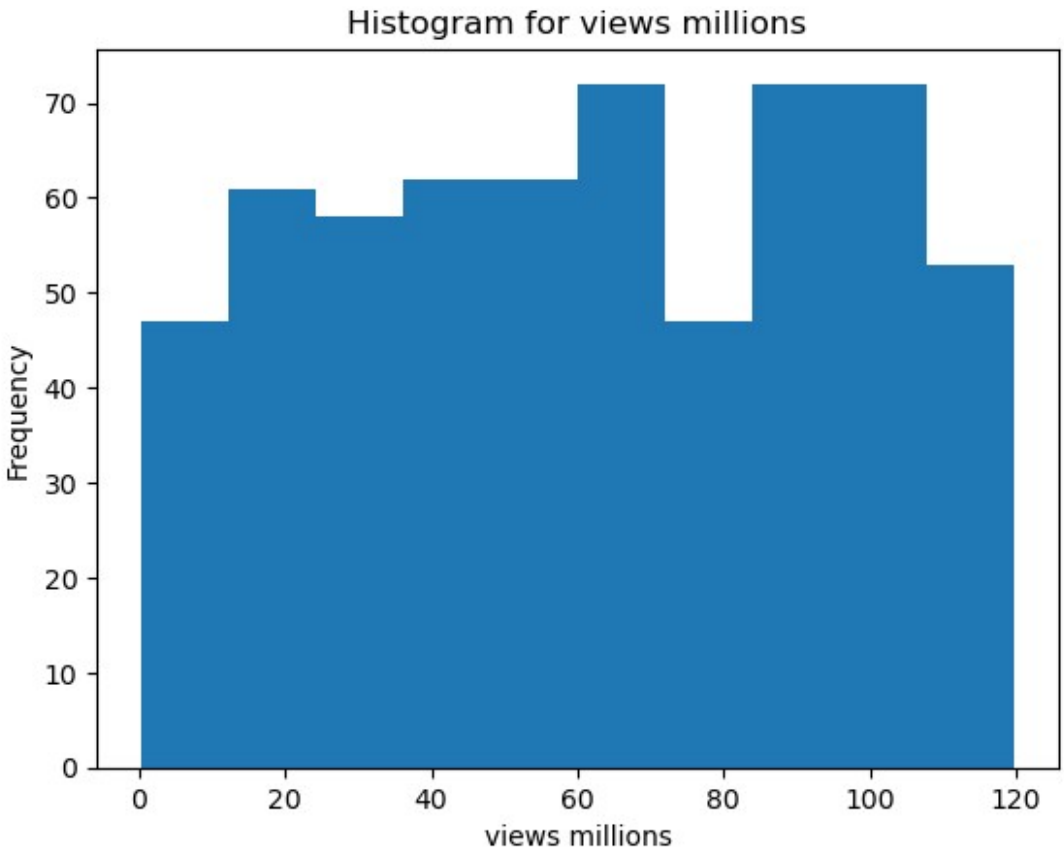
```
revenue_millions      float64
dtype: object
```

```
plt.hist(df["release_year"])
plt.title("Histogram for Release year")
plt.xlabel("Year")
plt.ylabel("Frequency")
plt.show()
```



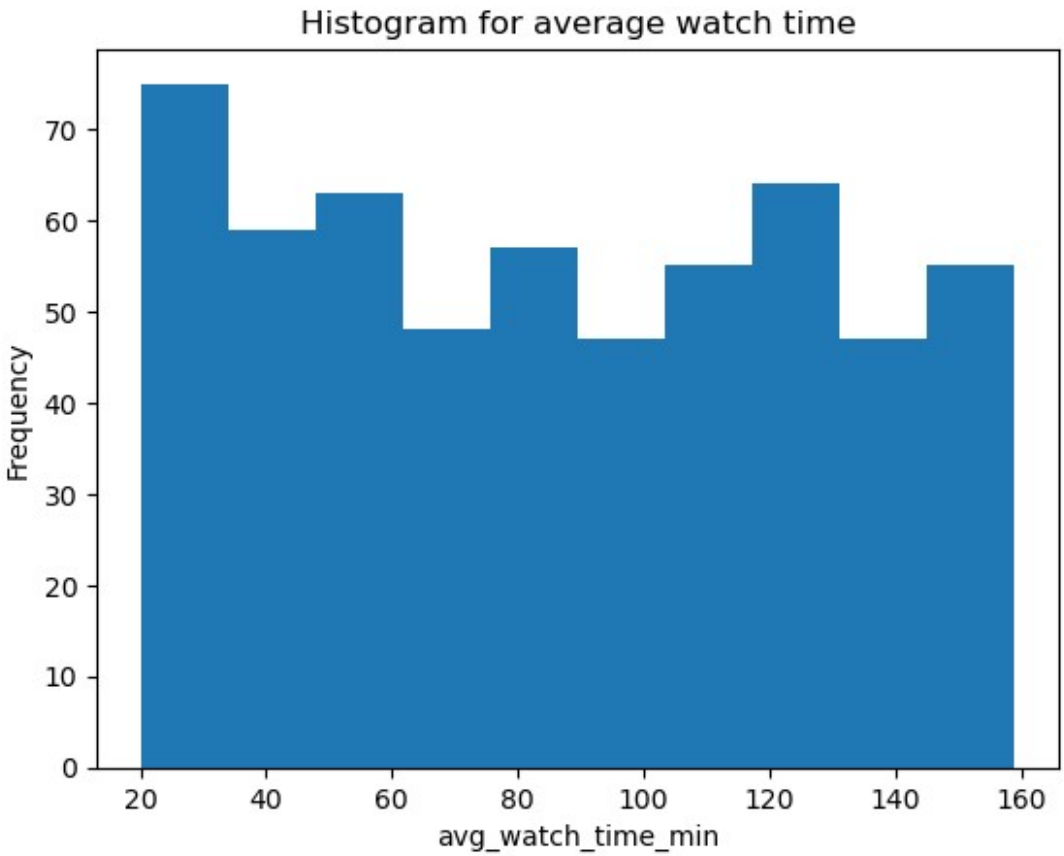
The distribution shows that higher sale in 2012-2017 year as follow 2022 and lowest sale in 2006-2010 year range.

```
plt.hist(df["views_millions"])
plt.title("Histogram for views millions")
plt.xlabel("views millions")
plt.ylabel("Frequency")
plt.show()
```



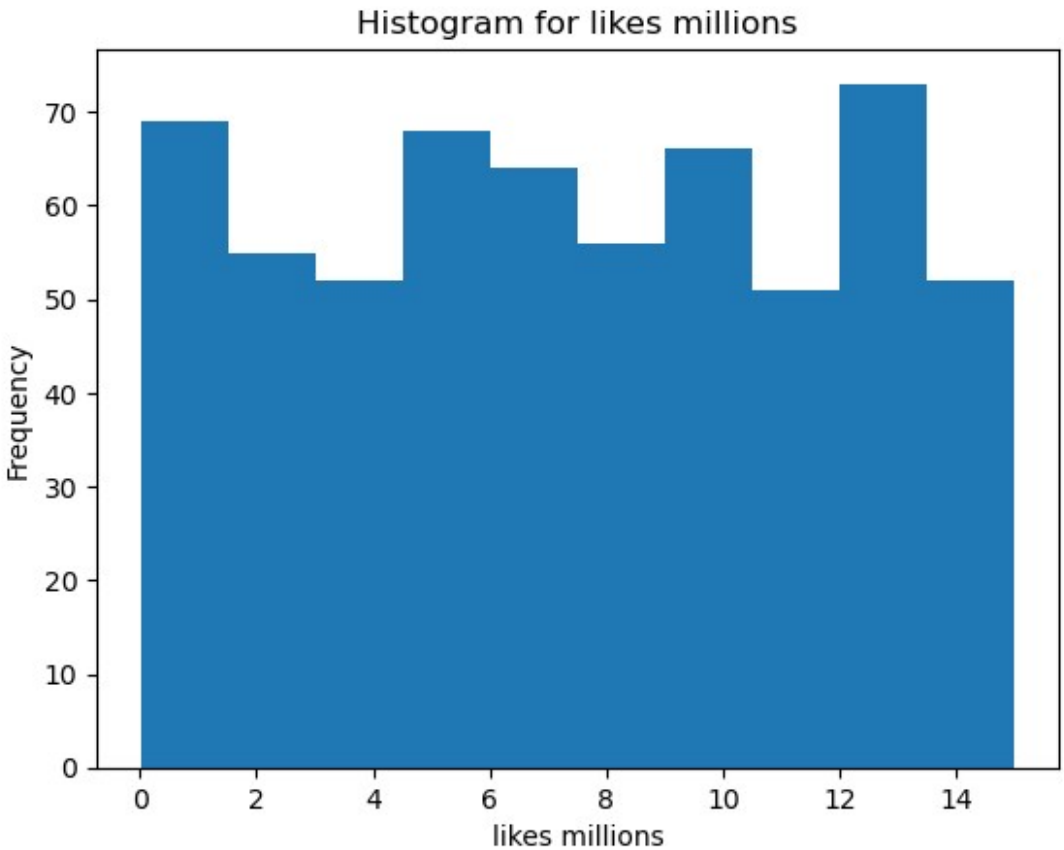
The distribution shows that the higher views are in range 60-70 and 80-110 and lower views are in range 0-15 as follow 70-80.

```
plt.hist(df["avg_watch_time_min"])
plt.title("Histogram for average watch time")
plt.xlabel("avg_watch_time_min")
plt.ylabel("Frequency")
plt.show()
```



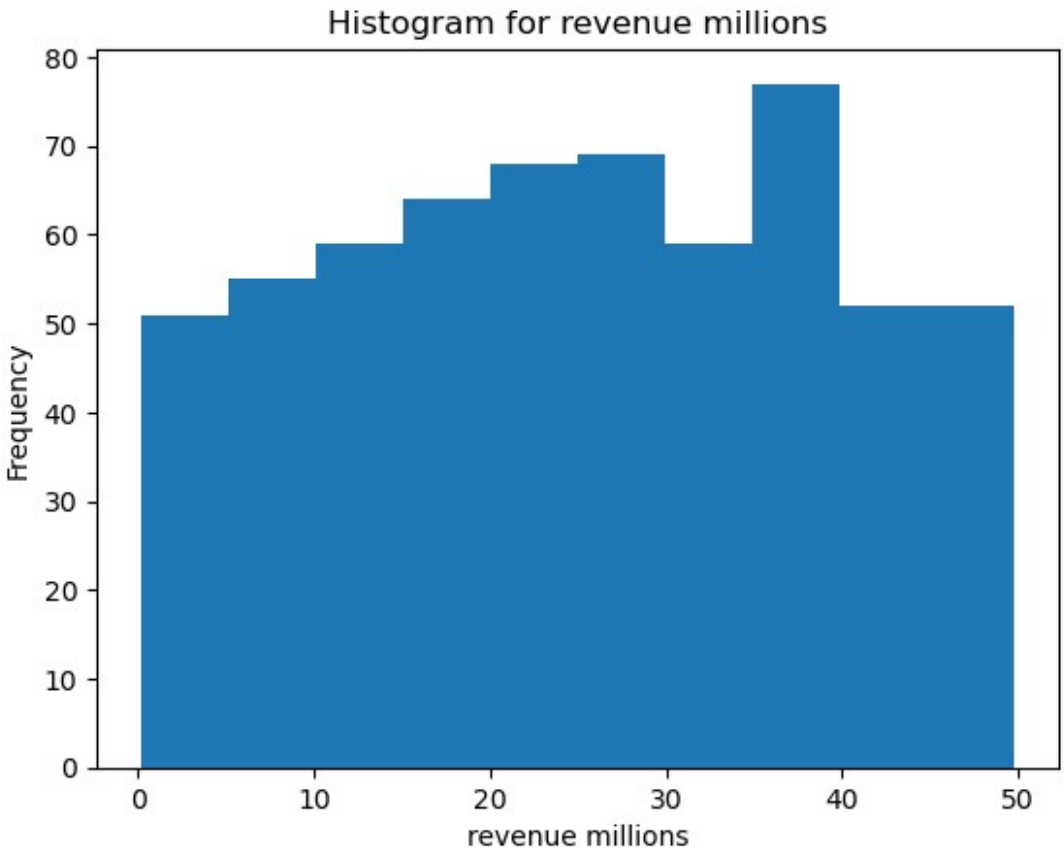
The distribution of average watch time in minutes shows that the higher watch time is 20-35 as follow 135-150 and lowest average watch time is 90-100 and 130-140.

```
plt.hist(df["likes_millions"])
plt.title("Histogram for likes millions")
plt.xlabel("likes millions")
plt.ylabel("Frequency")
plt.show()
```



The distribution of likes in millions shows that the higher likes in range 12-13.5 and lower is 10-12.

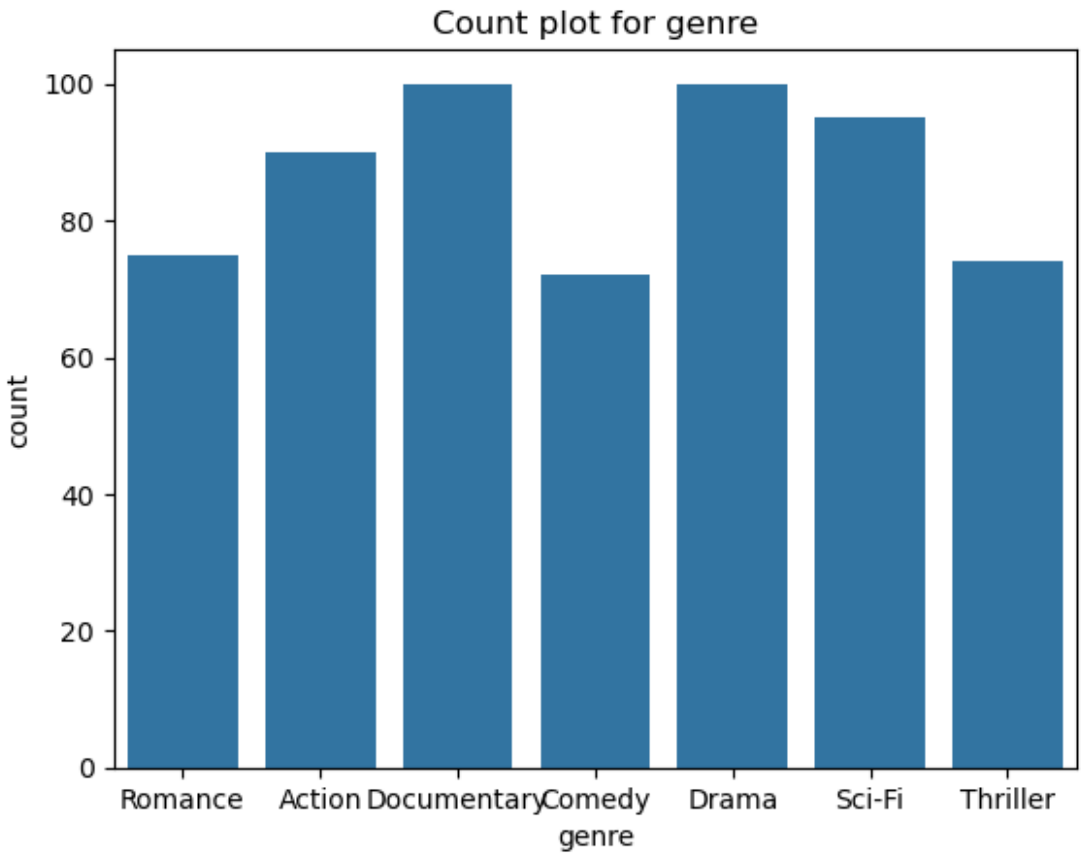
```
plt.hist(df["revenue_millions"])
plt.title("Histogram for revenue millions")
plt.xlabel("revenue millions")
plt.ylabel("Frequency")
plt.show()
```



The distribution of revenue in millions shows that the higher revenue in range 35-40 as follow 25-30 and lower revenue in range 0-5.

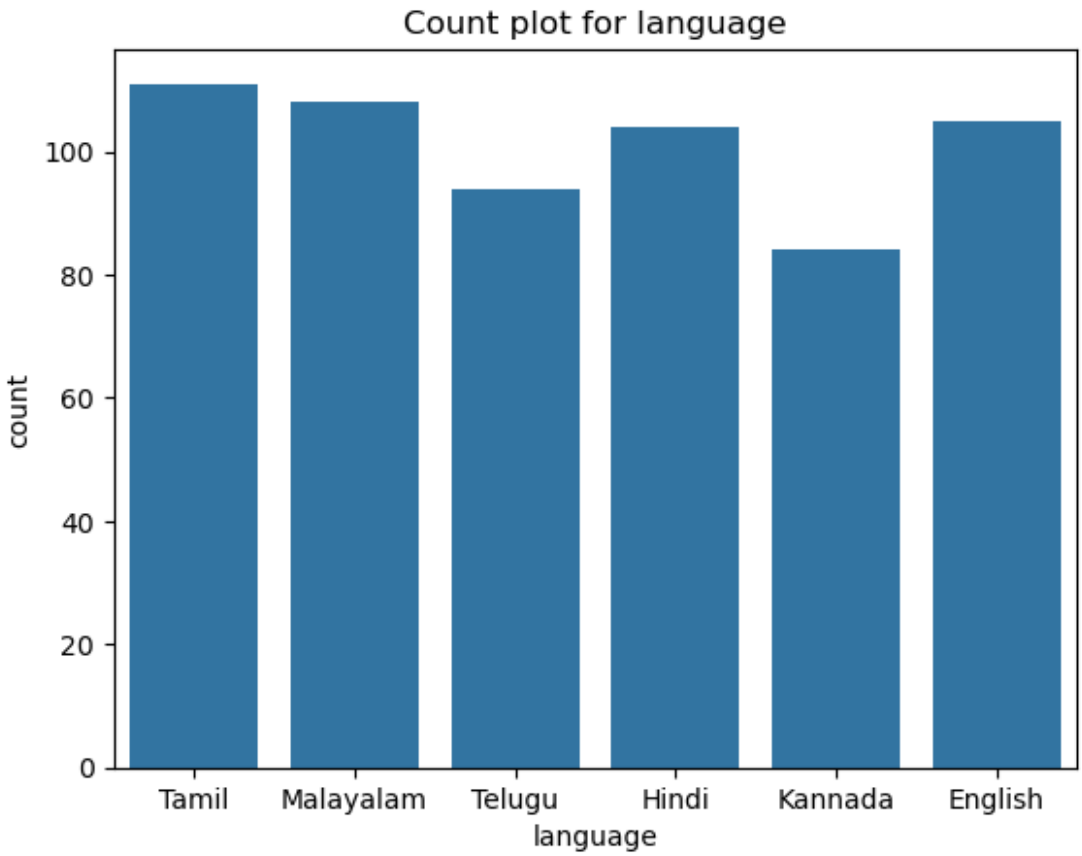
countplot

```
sns.countplot(x=df["genre"],data=df)
plt.title("Count plot for genre")
plt.xlabel("genre")
plt.show()
```



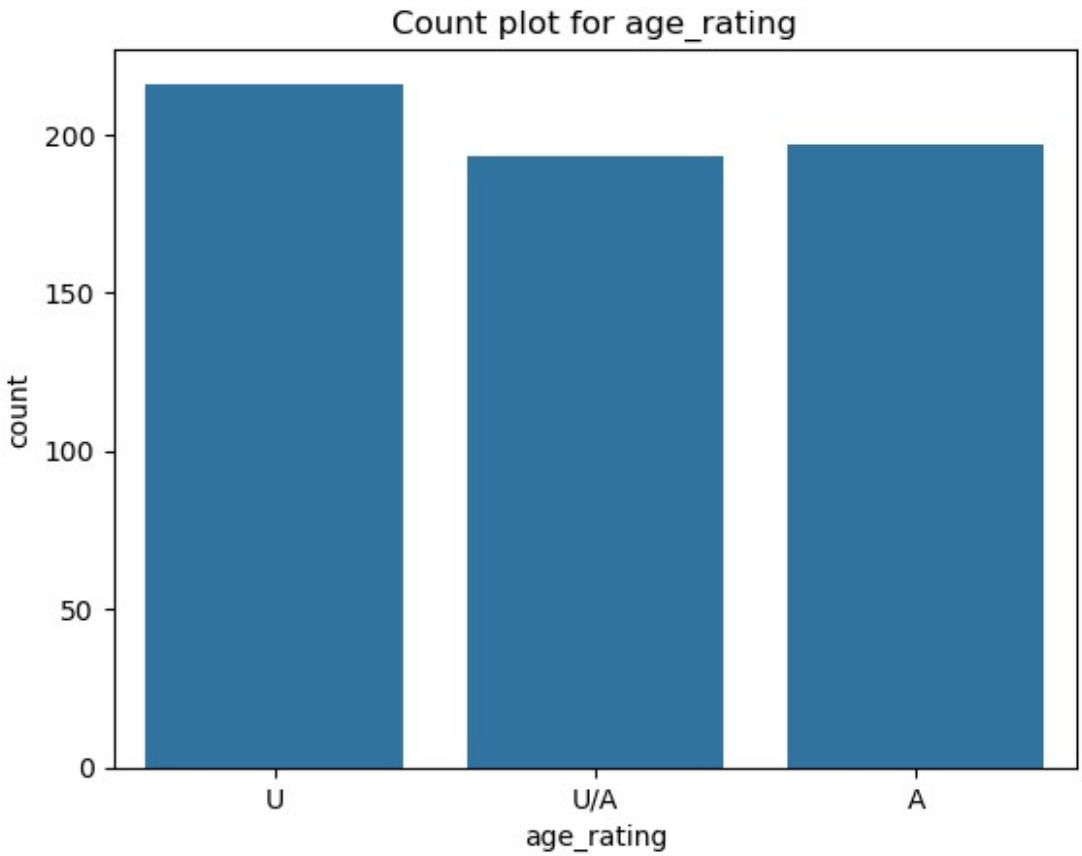
The count plot shows that the highest count of genre is documentary and drama and lowest count of genre is comedy as follow romance movies.

```
sns.countplot(x=df["language"],data=df)
plt.title("Count plot for language")
plt.xlabel("language")
plt.show()
```



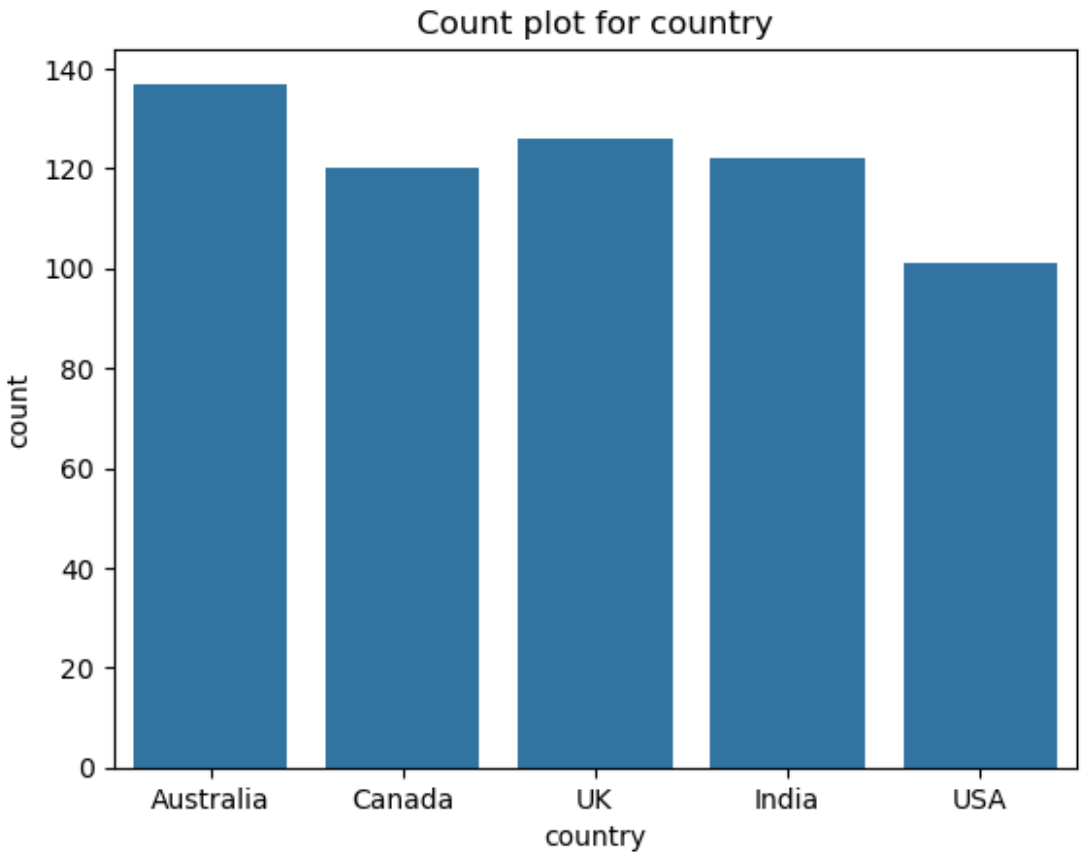
The count plot shows that the highest count of languages is tamil as follow malayalam and lowest count of language is kannada as follow telugu.

```
sns.countplot(x=df["age_rating"],data=df)
plt.title("Count plot for age_rating")
plt.xlabel("age_rating")
plt.show()
```



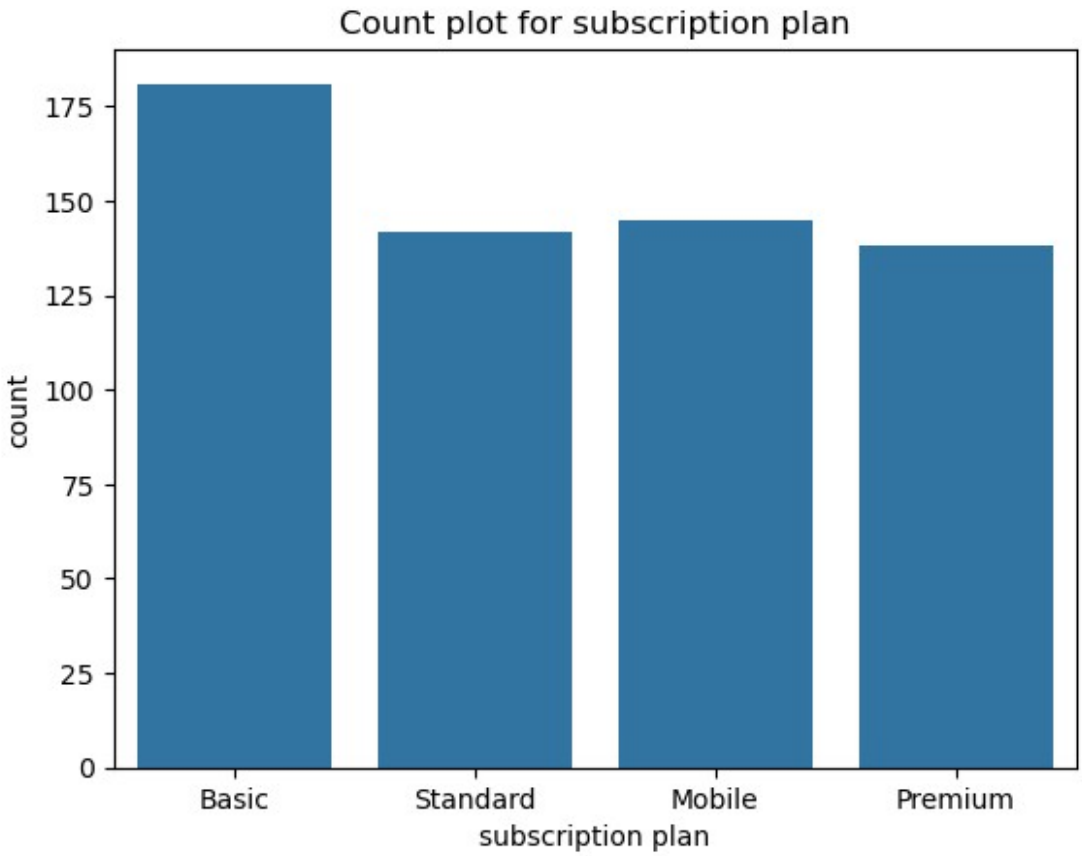
The count plot shows that the highest count of age rating is U and lowest count of age rating is U/A.

```
sns.countplot(x=df["country"],data=df)
plt.title("Count plot for country")
plt.xlabel("country")
plt.show()
```



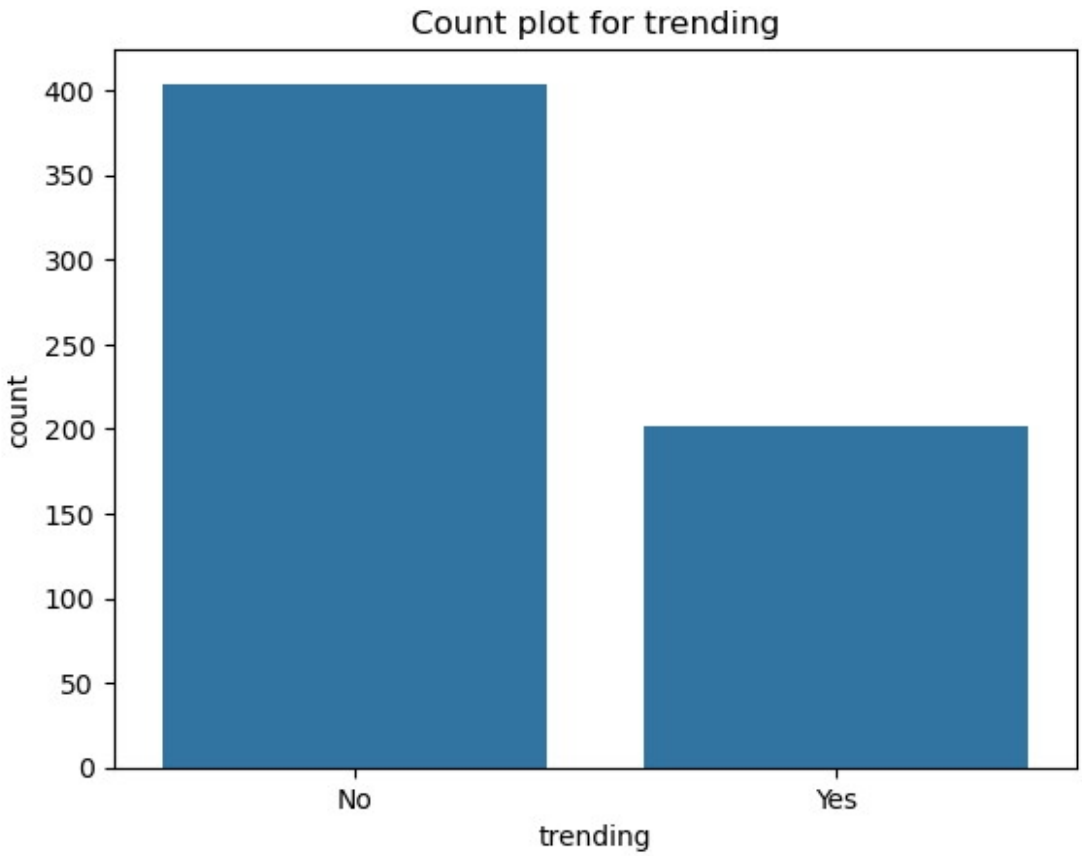
The count plot shows that the highest cout of country is Australia as follow UK and lowest count of country is USA as follow Canada.

```
sns.countplot(x=df["subscription_plan"],data=df)
plt.title("Count plot for subscription plan")
plt.xlabel("subscription plan")
plt.show()
```



The count plot shows that the highest count of subscription plan is Basic and lowest count of subscription plan is Premium.

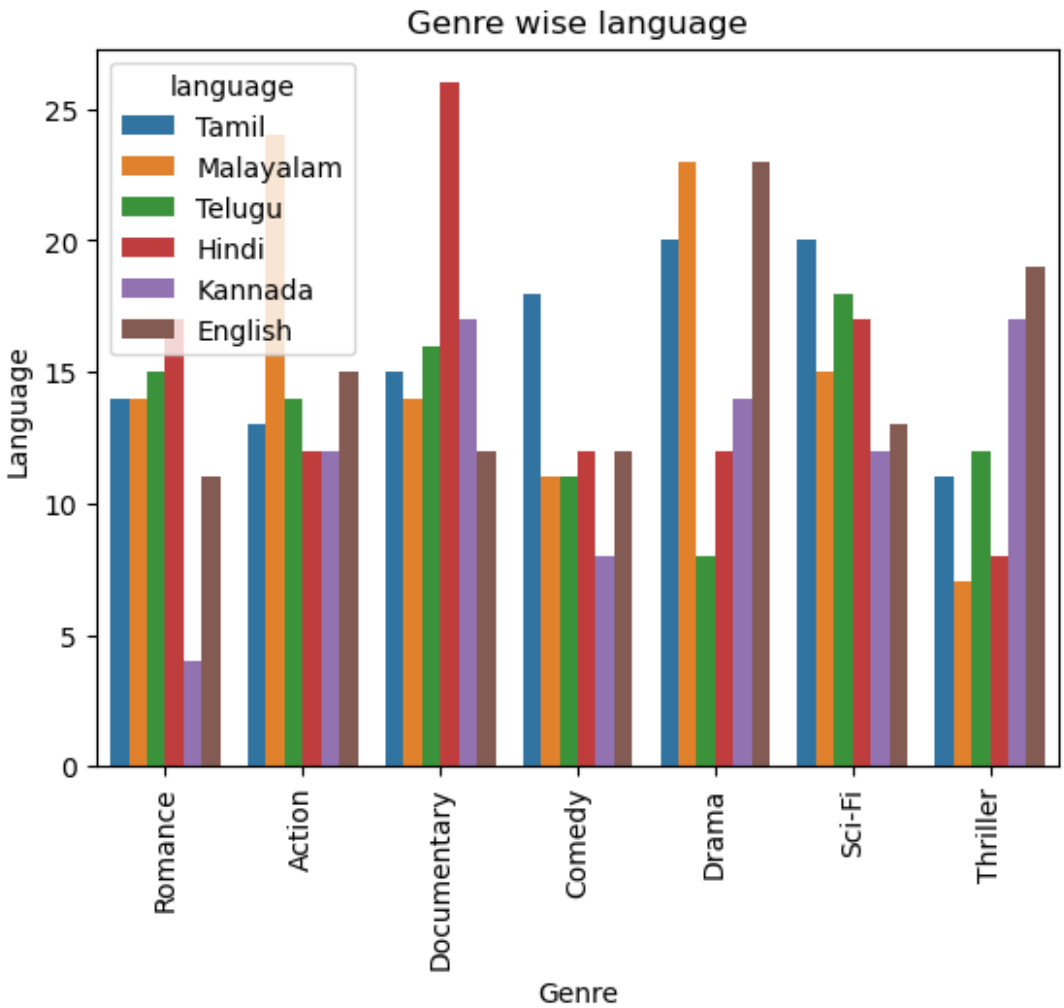
```
sns.countplot(x=df["is_trending"],data=df)
plt.title("Count plot for trending")
plt.xlabel("trending")
plt.show()
```



The count plot shows that the highest count of not trending and lowest is trending movies in Amazon prime.

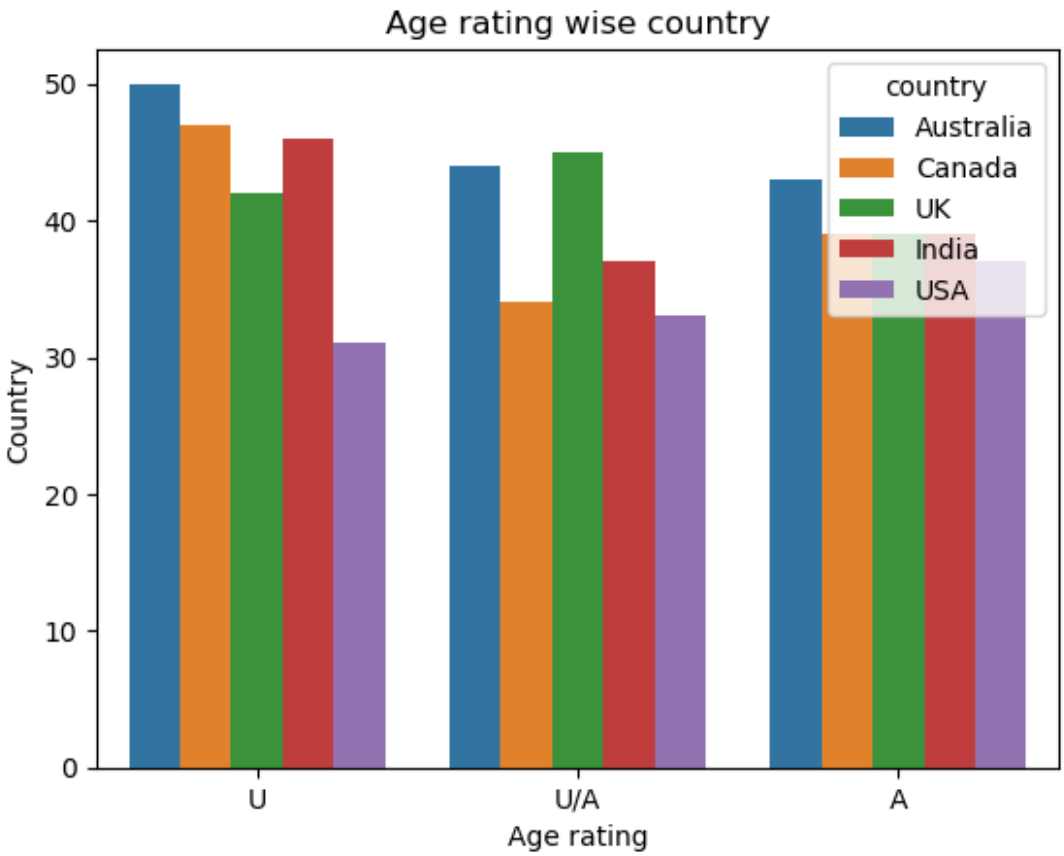
bivariate

```
sns.countplot(x=df["genre"],hue="language",data=df)
plt.title("Genre wise language")
plt.xlabel("Genre")
plt.ylabel("Language")
plt.xticks(rotation=90)
plt.show()
```



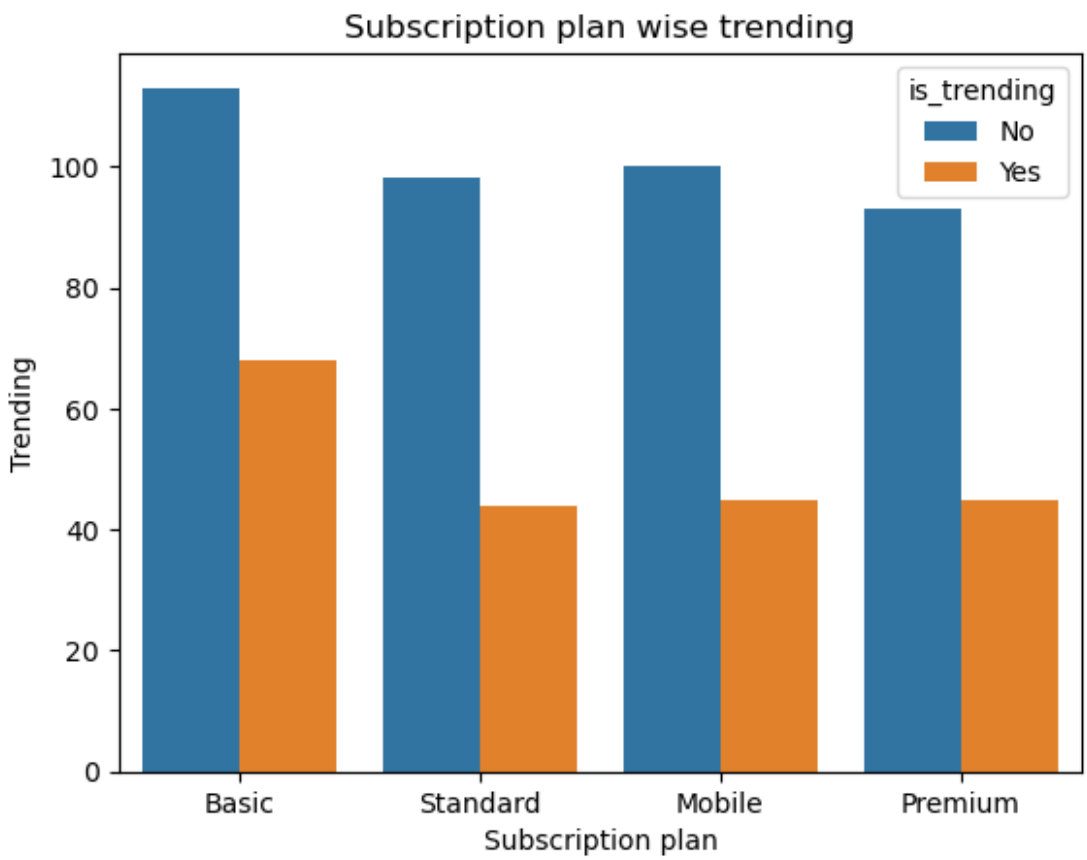
The Bivariate plot shows that the highest genre as per language is documentary in hindi language as follow action in hindi language and lowest genre as per language is romance in kannada language.

```
sns.countplot(x=df["age_rating"],hue="country",data=df)
plt.title("Age rating wise country")
plt.xlabel("Age rating")
plt.ylabel("Country")
plt.show()
```



The bivariate plot shows that the highest age rating as per country is U in Australia as follow canada and lowest age rating as per country is U in USA.

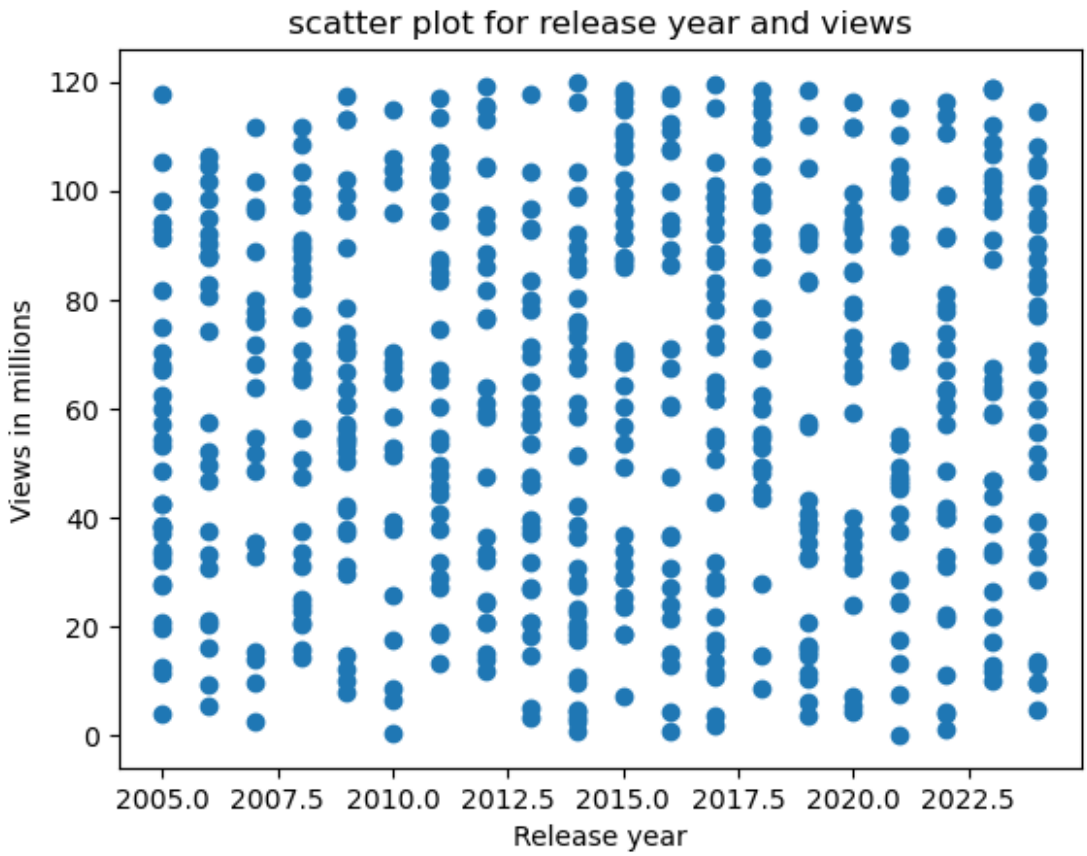
```
sns.countplot(x=df["subscription_plan"],hue="is_trending",data=df)
plt.title("Subscription plan wise trending")
plt.xlabel("Subscription plan")
plt.ylabel("Trending")
plt.show()
```



The bivariate plot shows that the highest subscription plan as per trending is Basic and lowest subscription plan as per trending is Standard.

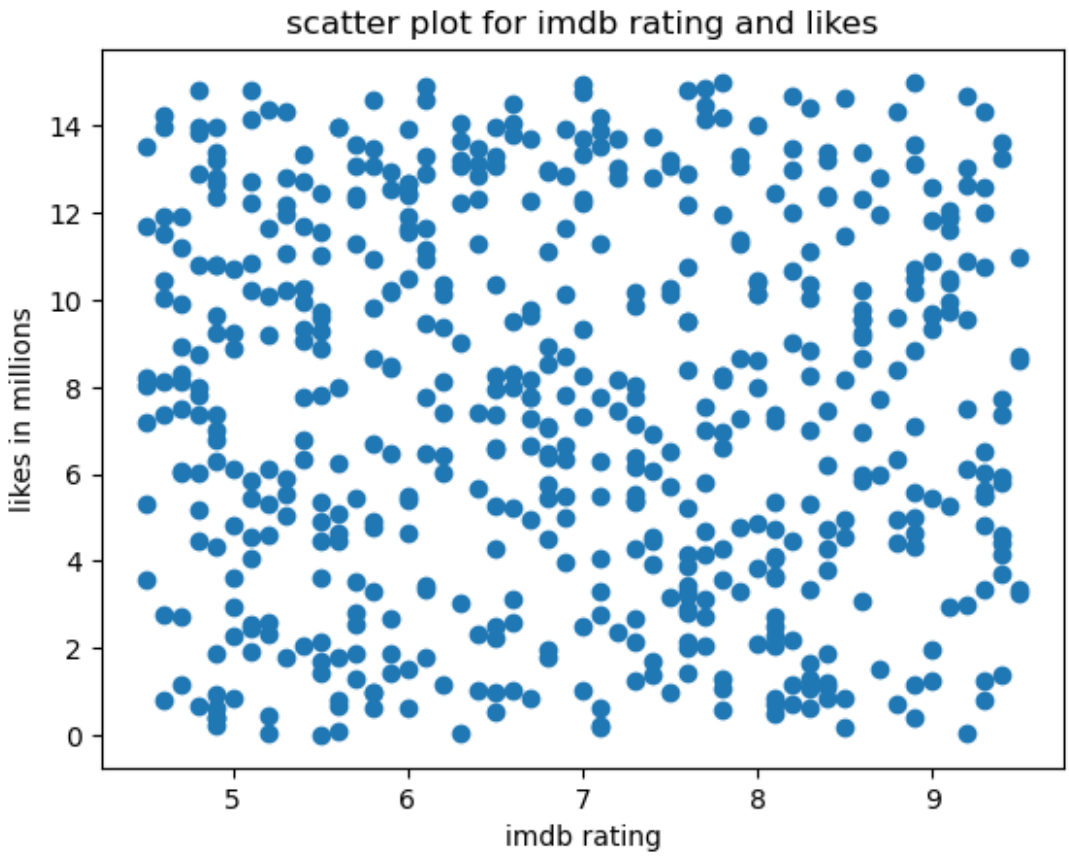
Scatter plot

```
plt.scatter(df["release_year"],df["views_millions"])
plt.title("scatter plot for release year and views")
plt.xlabel("Release year")
plt.ylabel("Views in millions")
plt.show()
```



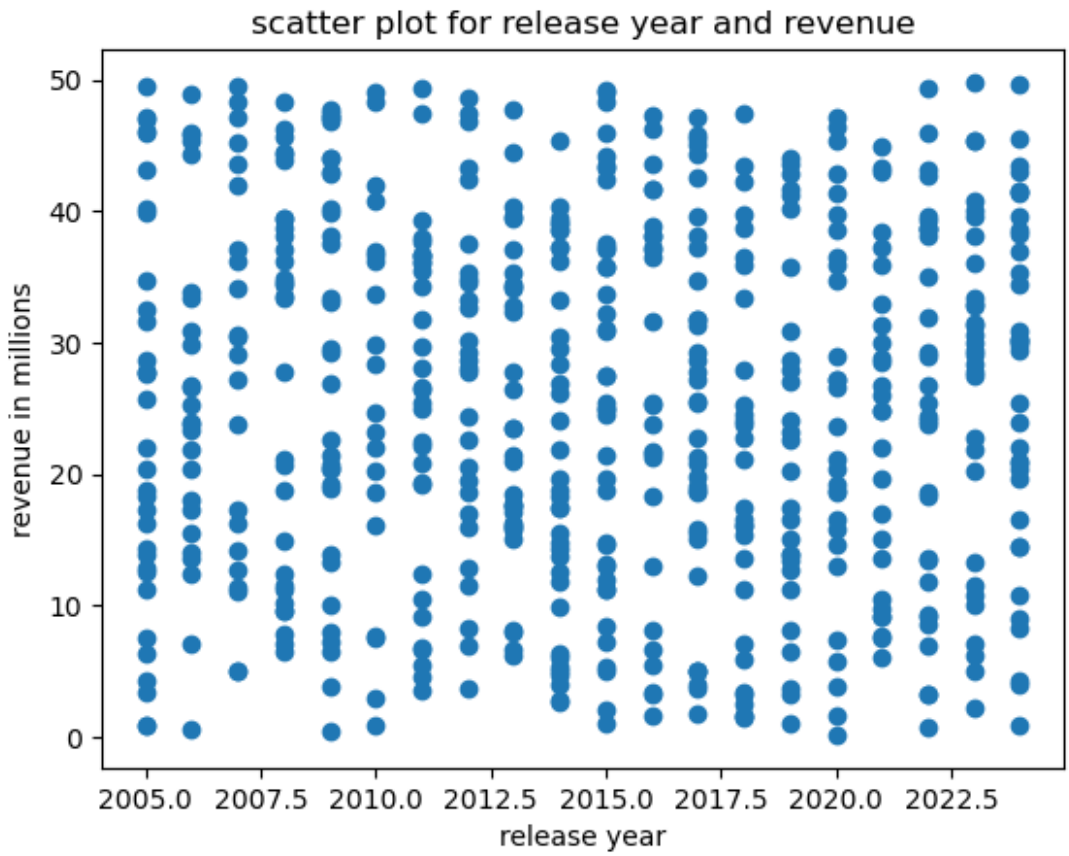
The scatter plot shows that the highest release year as per views is 2014-2017 and lowest is 2005-2010.

```
plt.scatter(df["imdb_rating"],df["likes_millions"])
plt.title("scatter plot for imdb rating and likes")
plt.xlabel("imdb rating")
plt.ylabel("likes in millions")
plt.show()
```



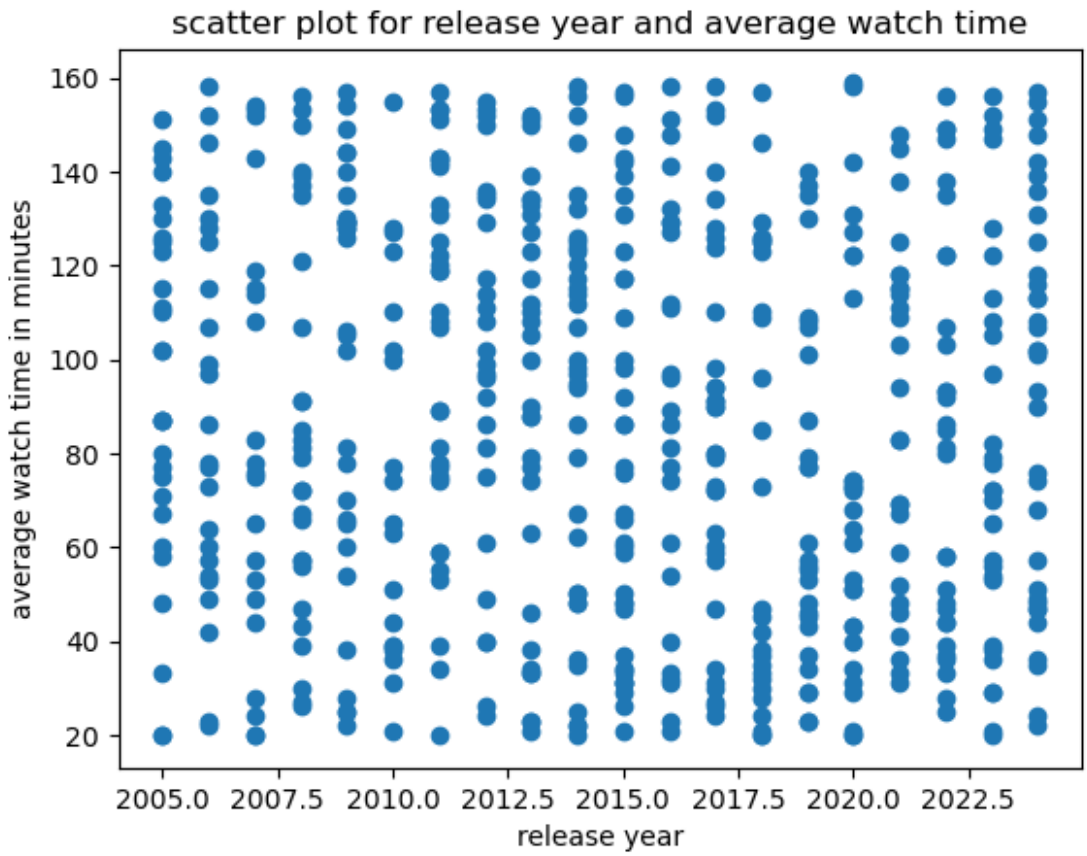
The scatter plot shows that there is no strong relationship between imdb rating and likes.It is very sometime imdb rating high then likes few or imdb rating low the likes higher in this plot it doesn't form any line.

```
plt.scatter(df["release_year"],df["revenue_millions"])
plt.title("scatter plot for release year and revenue")
plt.xlabel("release year")
plt.ylabel("revenue in millions")
plt.show()
```



The scatter plot shows that the higher revenue in releasing year is 2015 and lowest is 2018.There some values who gets the higher or lower revenue in those years.

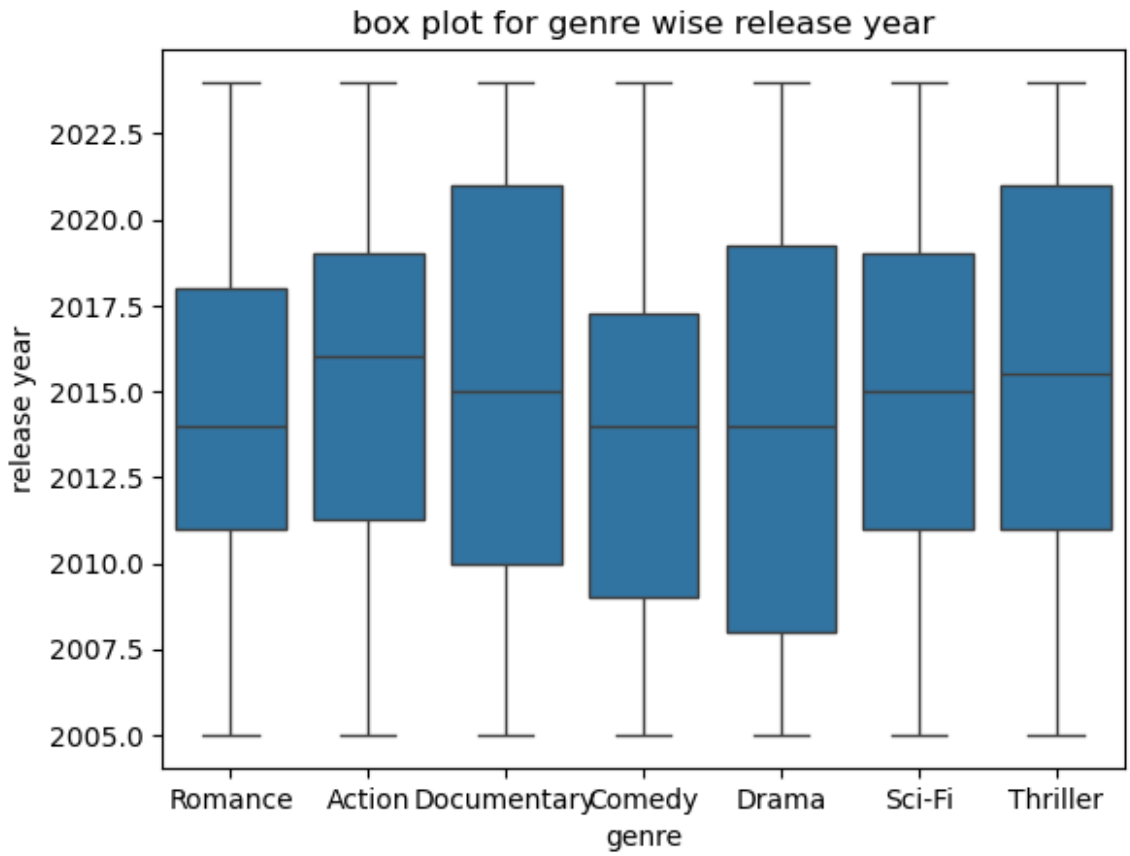
```
plt.scatter(df["release_year"],df["avg_watch_time_min"])
plt.title("scatter plot for release year and average watch time")
plt.xlabel("release year")
plt.ylabel("average watch time in minutes")
plt.show()
```



The scatter plot shows that the higher average watch time in releasing year is 2012-2015 and lower 2017-2020.

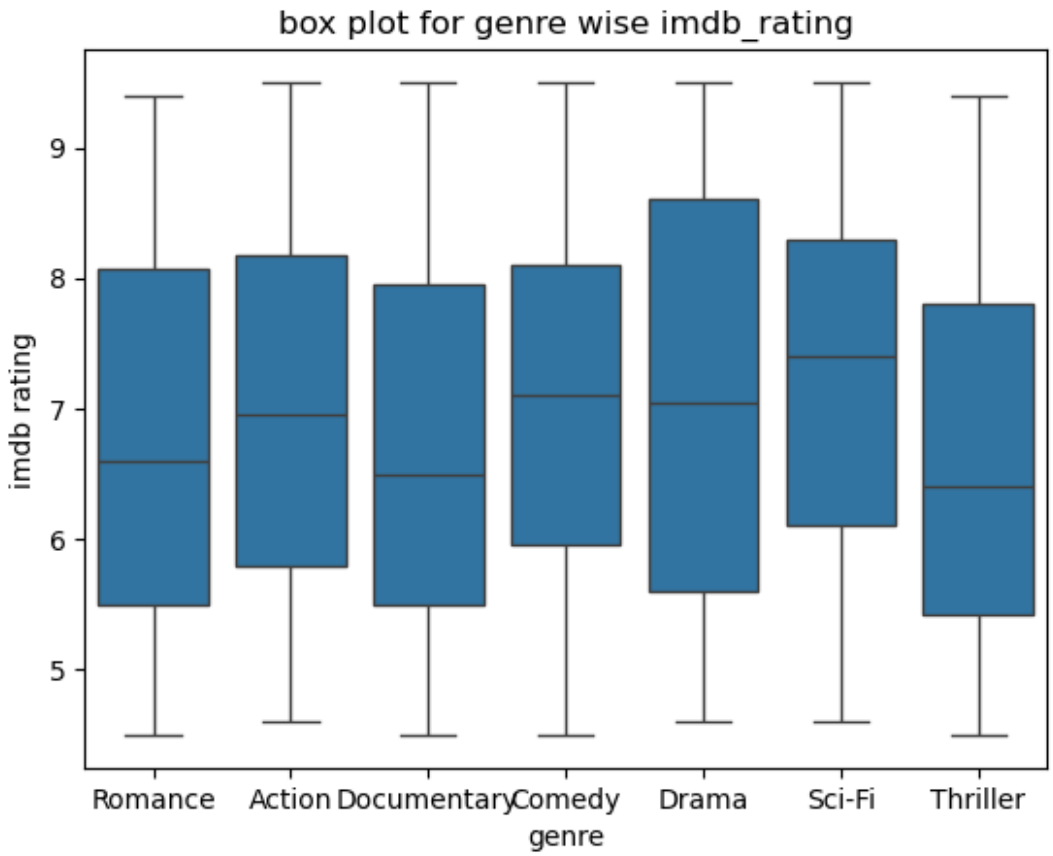
Box plot

```
sns.boxplot(x="genre",y="release_year",data=df)
plt.title("box plot for genre wise release year")
plt.xlabel("genre")
plt.ylabel("release year")
plt.show()
```



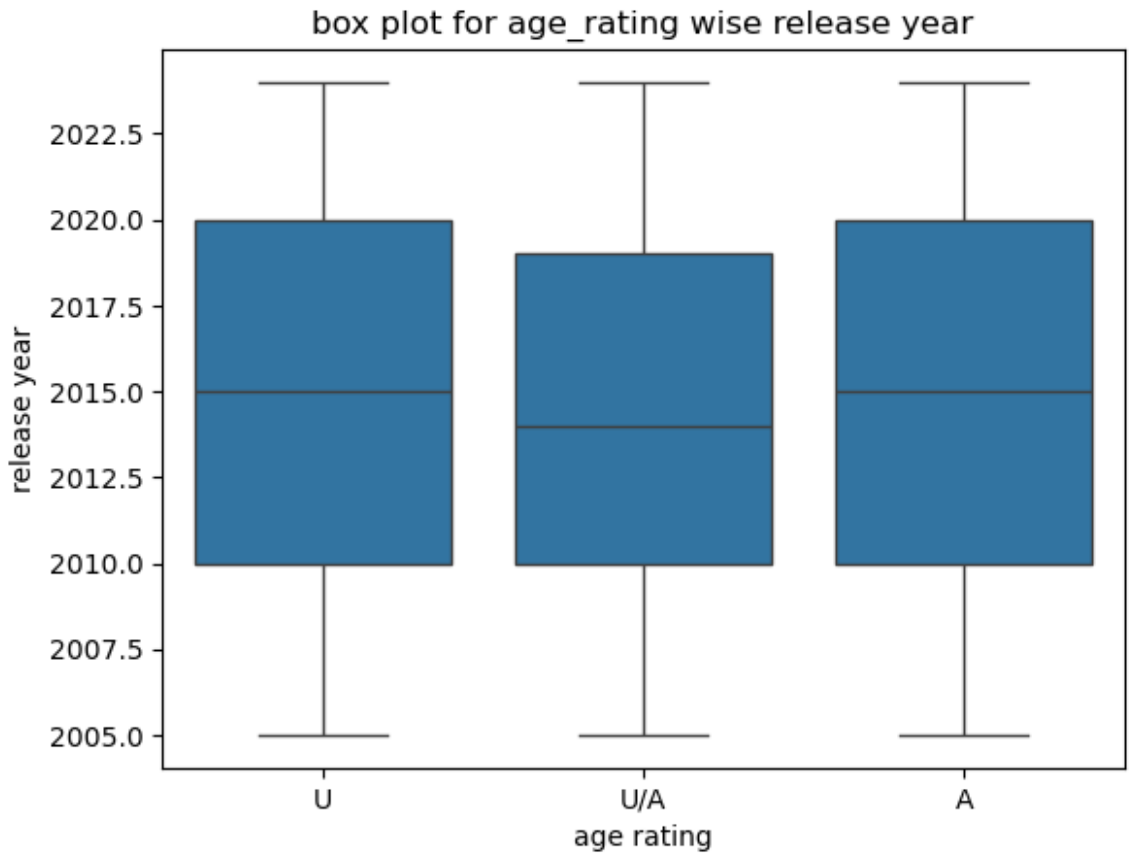
The box plot shows that the highest median of genre as per release year is drama and lowest median is romance.

```
sns.boxplot(x="genre",y="imdb_rating",data=df)
plt.title("box plot for genre wise imdb_rating")
plt.xlabel("genre")
plt.ylabel("imdb rating")
plt.show()
```



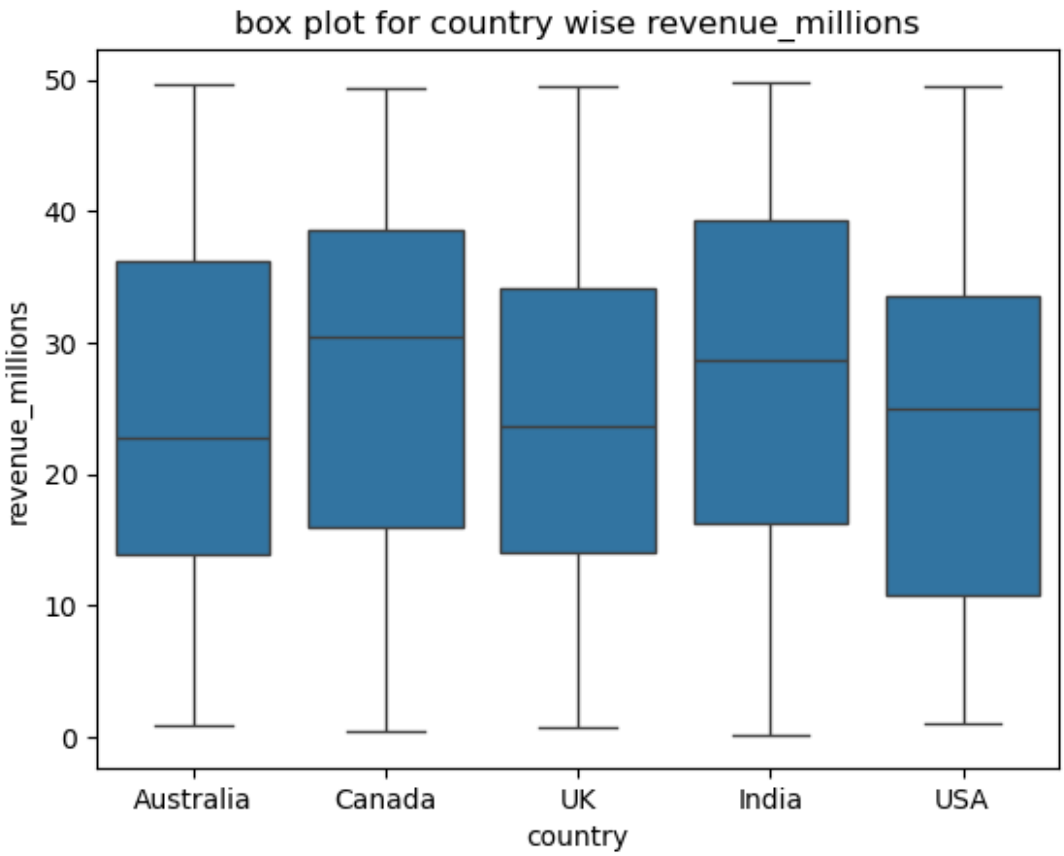
the box plot shows that the higher median of genre as per imdb rating is comedy as follow sci-fi and lowest median is thriller and documentary.

```
sns.boxplot(x="age_rating",y="release_year",data=df)
plt.title("box plot for age_rating wise release year")
plt.xlabel("age rating")
plt.ylabel("release year")
plt.show()
```



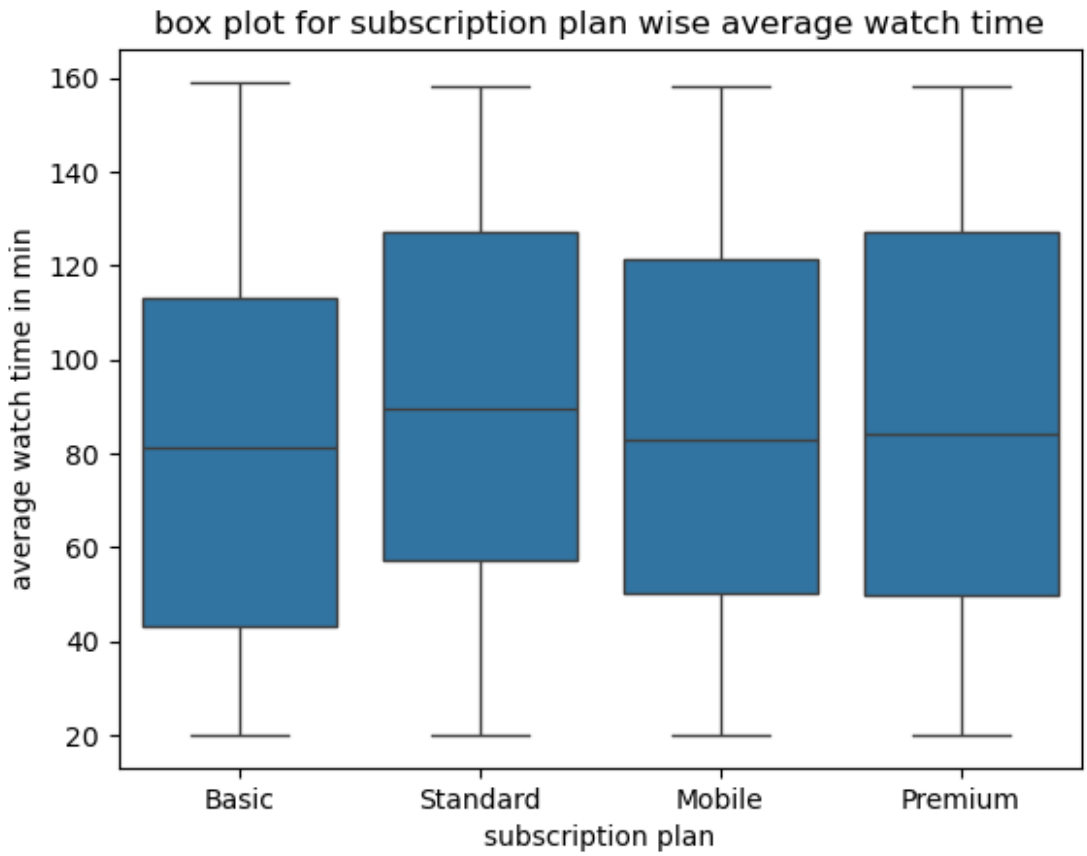
the box plot shows that the highest median of age rating as per release year is A as follow U and lowest median is U/A.

```
sns.boxplot(x="country",y="revenue_millions",data=df)
plt.title("box plot for country wise revenue_millions")
plt.xlabel("country")
plt.ylabel("revenue_millions")
plt.show()
```



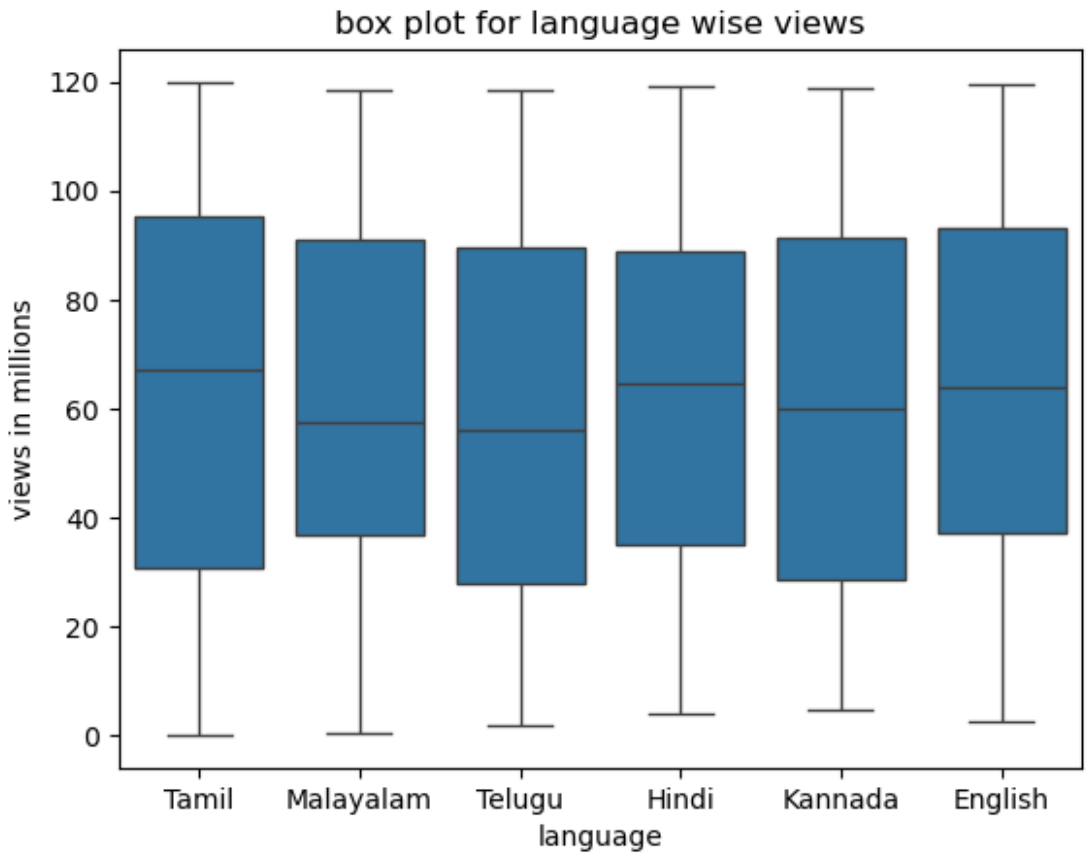
the box plot shows that the highest median country as per revenue is canada and USA and lowest median is Australia.

```
sns.boxplot(x="subscription_plan",y="avg_watch_time_min",data=df)
plt.title("box plot for subscription plan wise average watch time")
plt.xlabel("subscription plan")
plt.ylabel("average watch time in min")
plt.show()
```



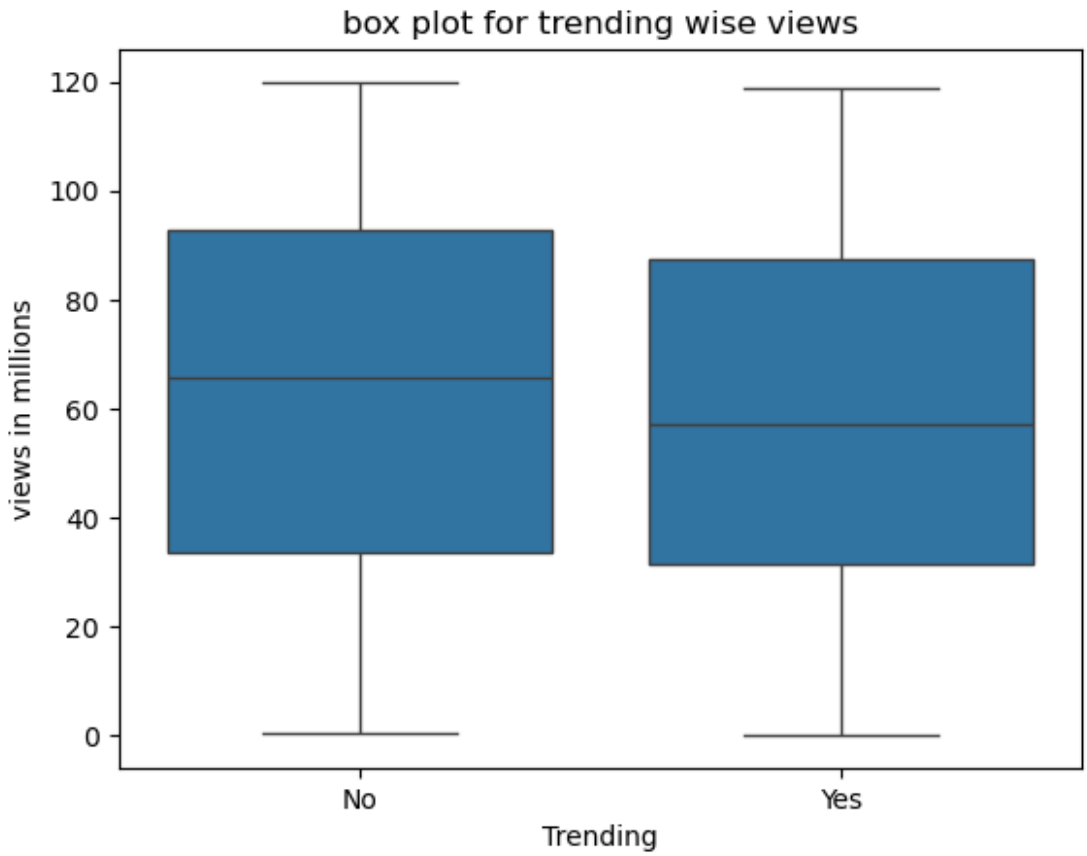
the box plot shows that the highest median of subscription plan as per average watch time is basic and lowest median is standard.

```
sns.boxplot(x="language",y="views_millions",data=df)
plt.title("box plot for language wise views")
plt.xlabel("language")
plt.ylabel("views in millions")
plt.show()
```



the box plot shows that the highest median of language as per views is tamil as follow hindi and lowest median is malayalam as follow telugu.

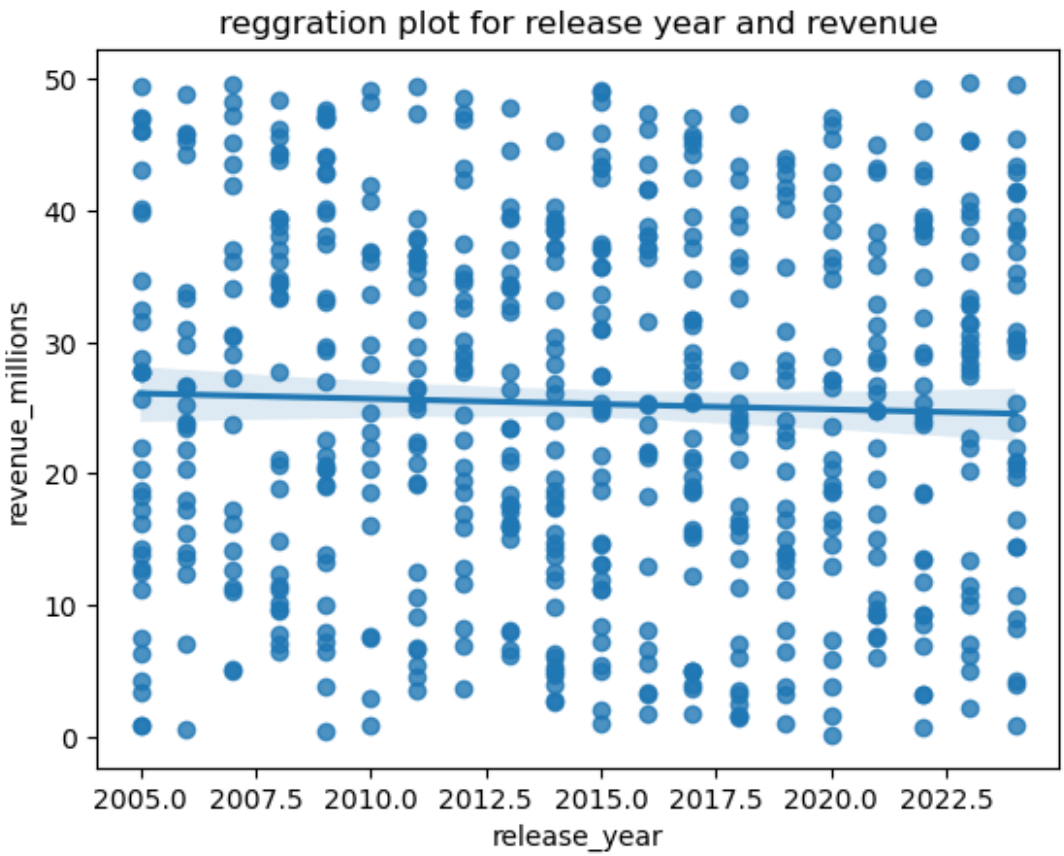
```
sns.boxplot(x="is_trending",y="views_millions",data=df)
plt.title("box plot for trending wise views")
plt.xlabel("Trending")
plt.ylabel("views in millions")
plt.show()
```



the box plot shows that the highest median of trending as per views is not trending and lowest median is trending.

Reggration plot

```
sns.regplot(x="release_year",y="revenue_millions",data=df)
plt.title("reggration plot for release year and revenue")
plt.show()
```

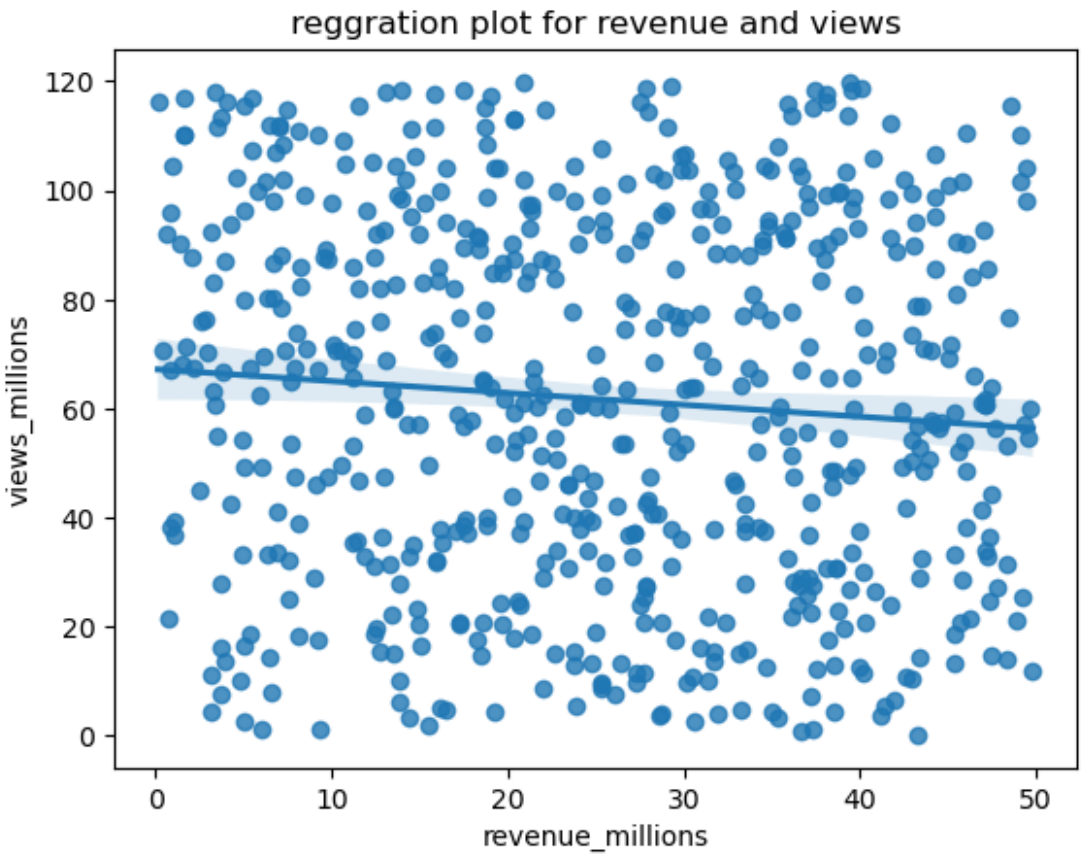


the reggration plot shows that the downward trend in revenue as per release year.That means revenue decreases as per low releasing movies as per up coming year.

```
sns.regplot(x="imdb_rating",y="revenue_millions",data=df)
plt.title("reggration plot for imdb rating and revenue")
plt.show()
```

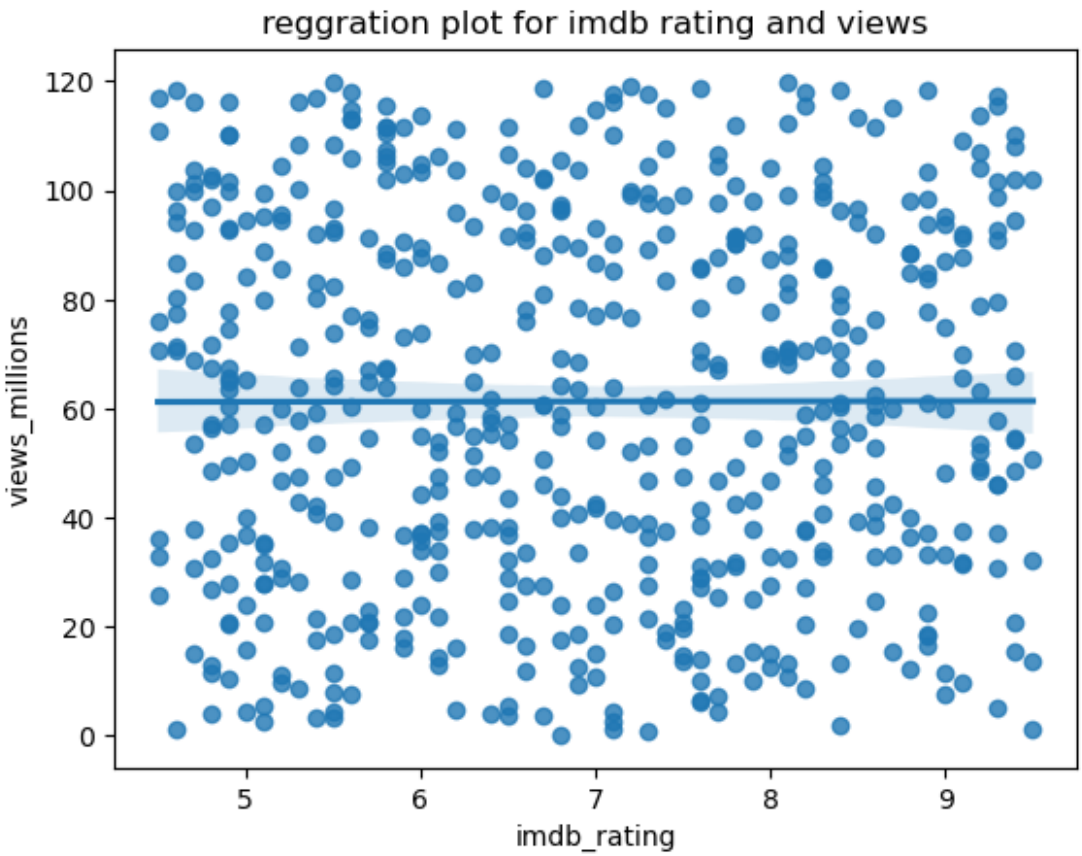
the reggration plot shows that the upward trend in revenue as per imdb rating.That means the imdb rating is increases then revenue also increases.

```
sns.regplot(x="revenue_millions",y="views_millions",data=df)
plt.title("reggration plot for revenue and views")
plt.show()
```



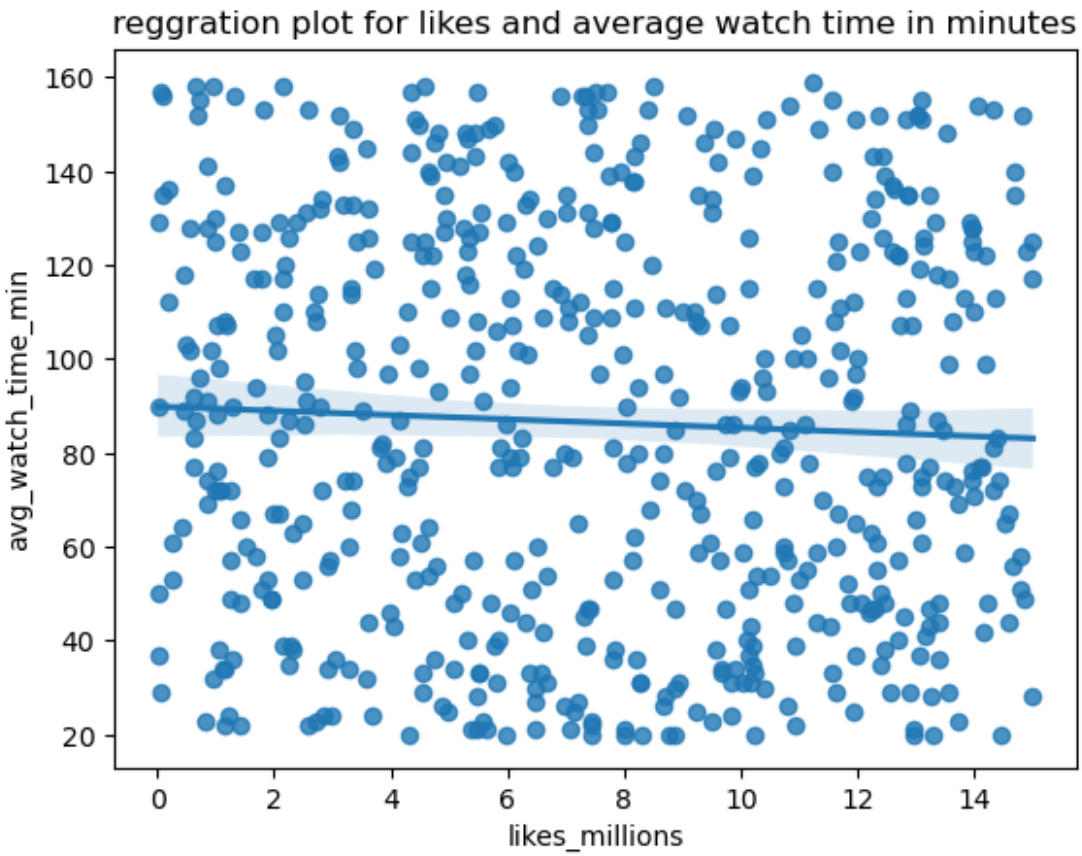
the reggration plot shows that the downward trend in revenue as per views. that means revenue increases then views are decreases.

```
sns.regplot(x="imdb_rating",y="views_millions",data=df)
plt.title("reggration plot for imdb rating and views")
plt.show()
```



the reggration plot shows that there is uniform trend in imdb rating as per viwes.

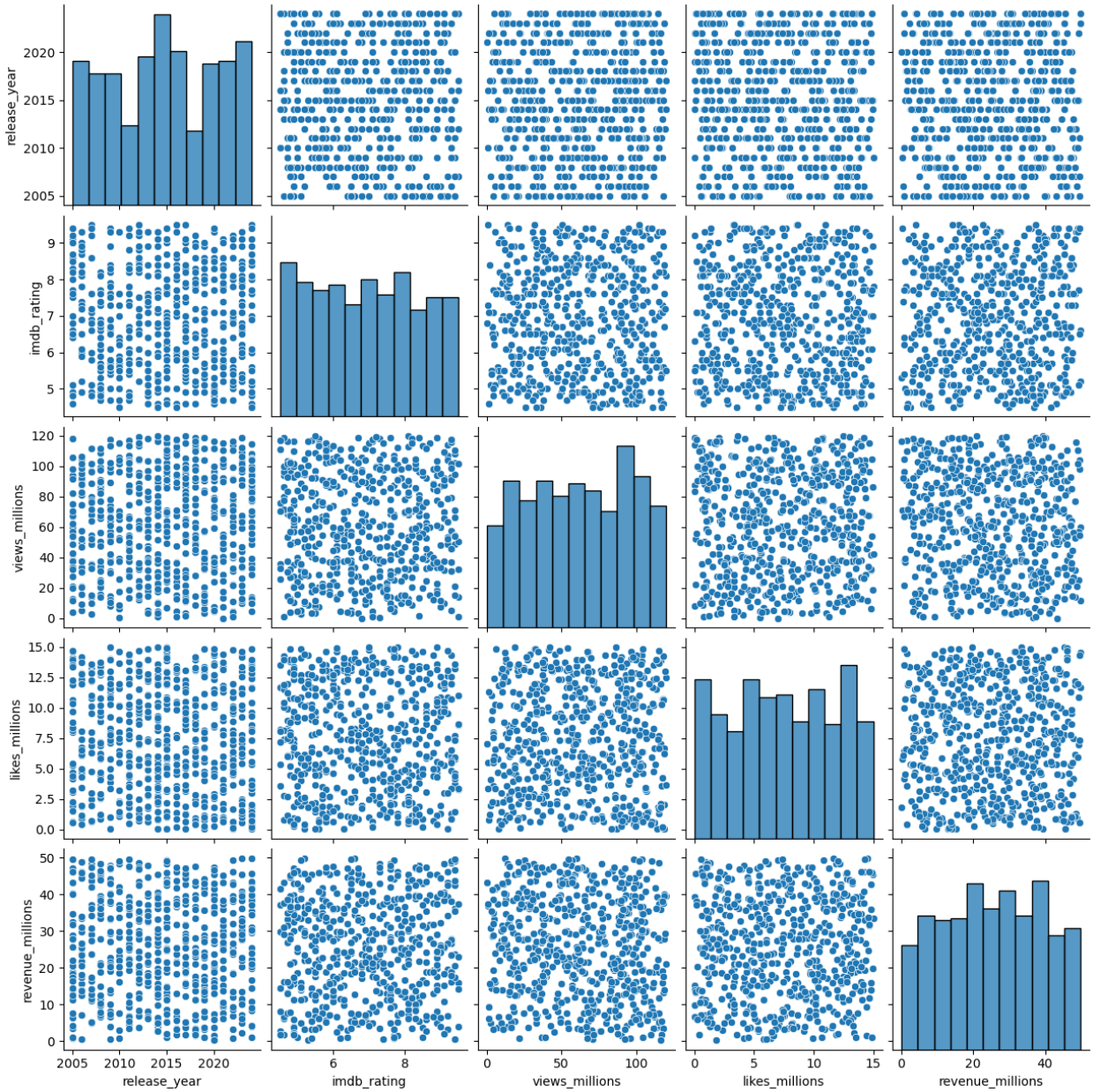
```
sns.regplot(x="likes_millions",y="avg_watch_time_min",data=df)
plt.title("reggration plot for likes and average watch time in
minutes")
plt.show()
```



the reggration plot shows that the downward trend in likes as per average watch time.That means when likes increases then average watch time is decreasing.

Pair plot

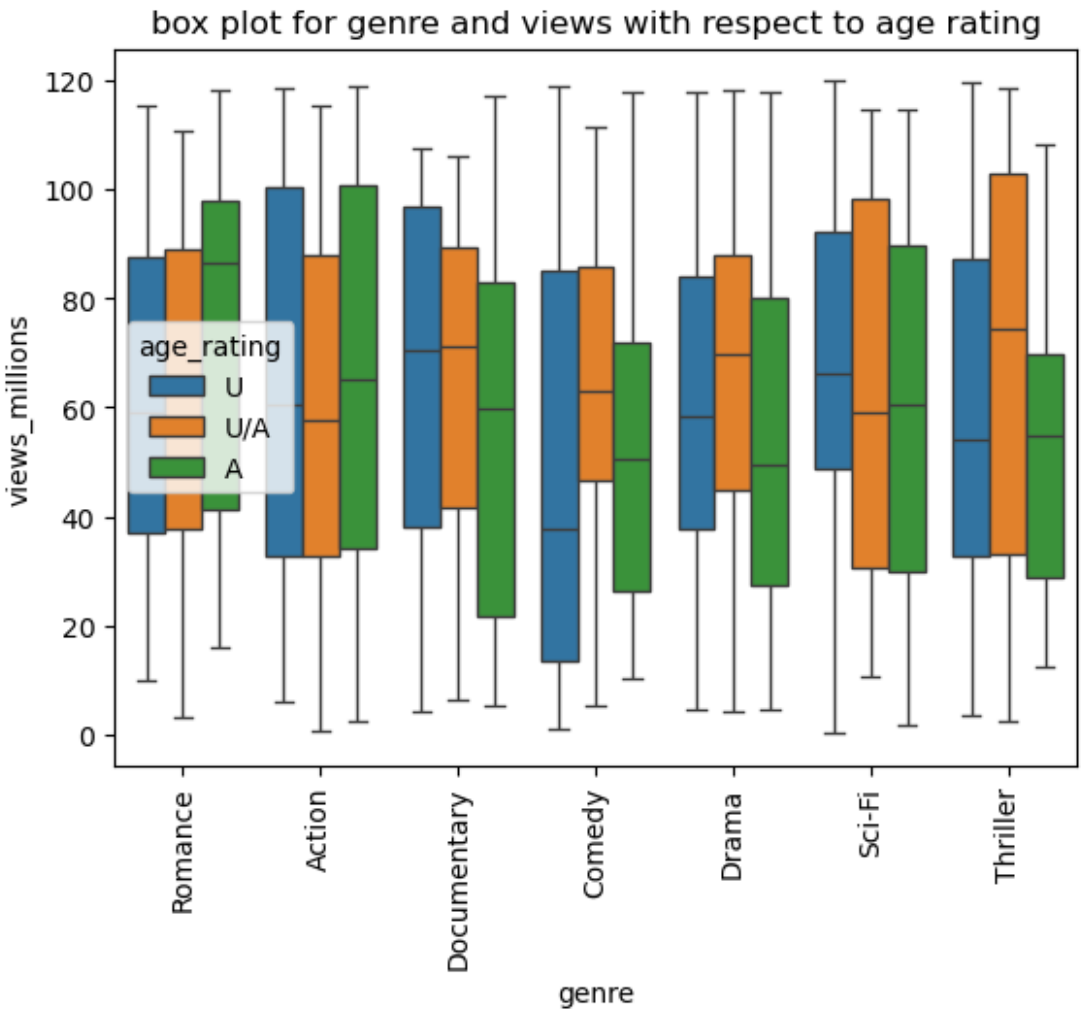
```
sns.pairplot(df[["release_year","imdb_rating","views_millions","likes_
millions","revenue_millions"]])
plt.show()
```



Through pair plot there is no strong relationship between in these columns.There is no proper line form to show details of the data.

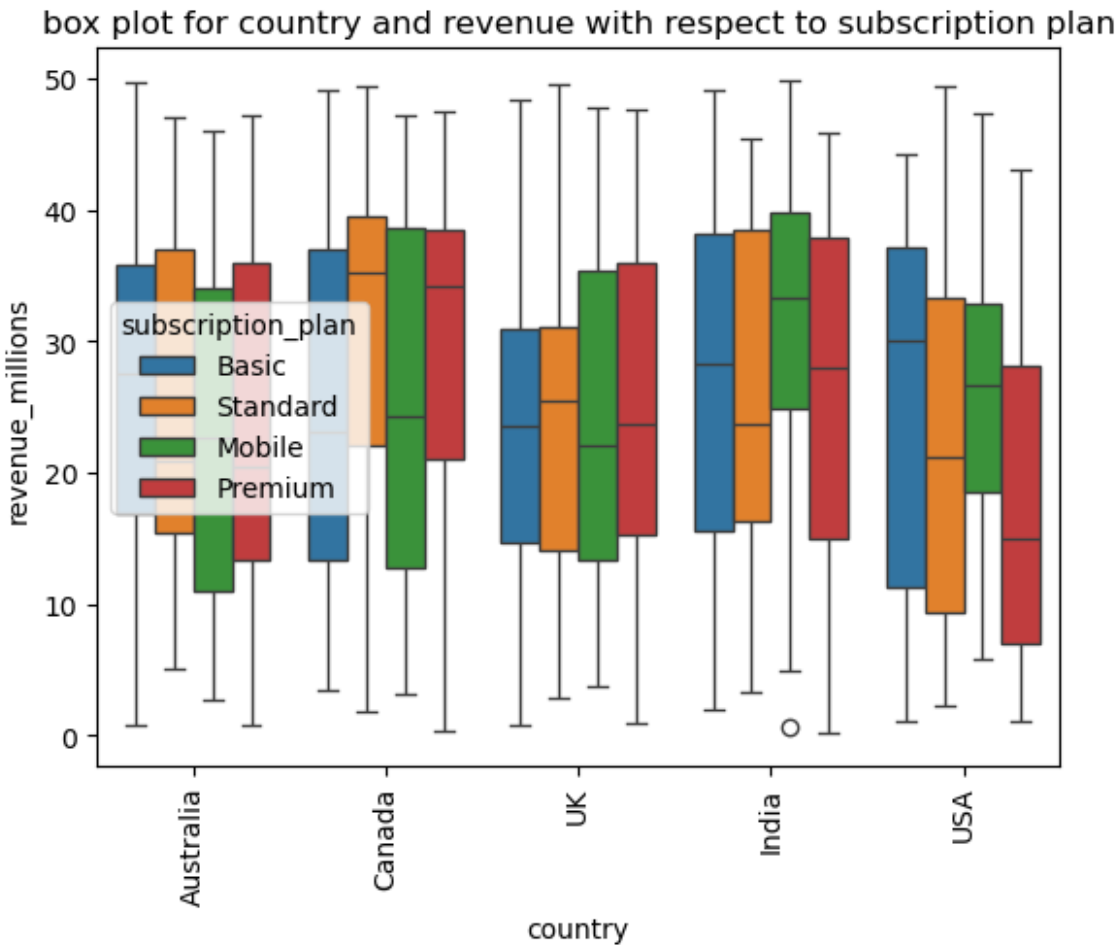
Box plot(bivariate)

```
sns.boxplot(x="genre",y="views_millions",hue="age_rating",data=df)
plt.title("box plot for genre and views with respect to age rating")
plt.xticks(rotation =90)
plt.show()
```



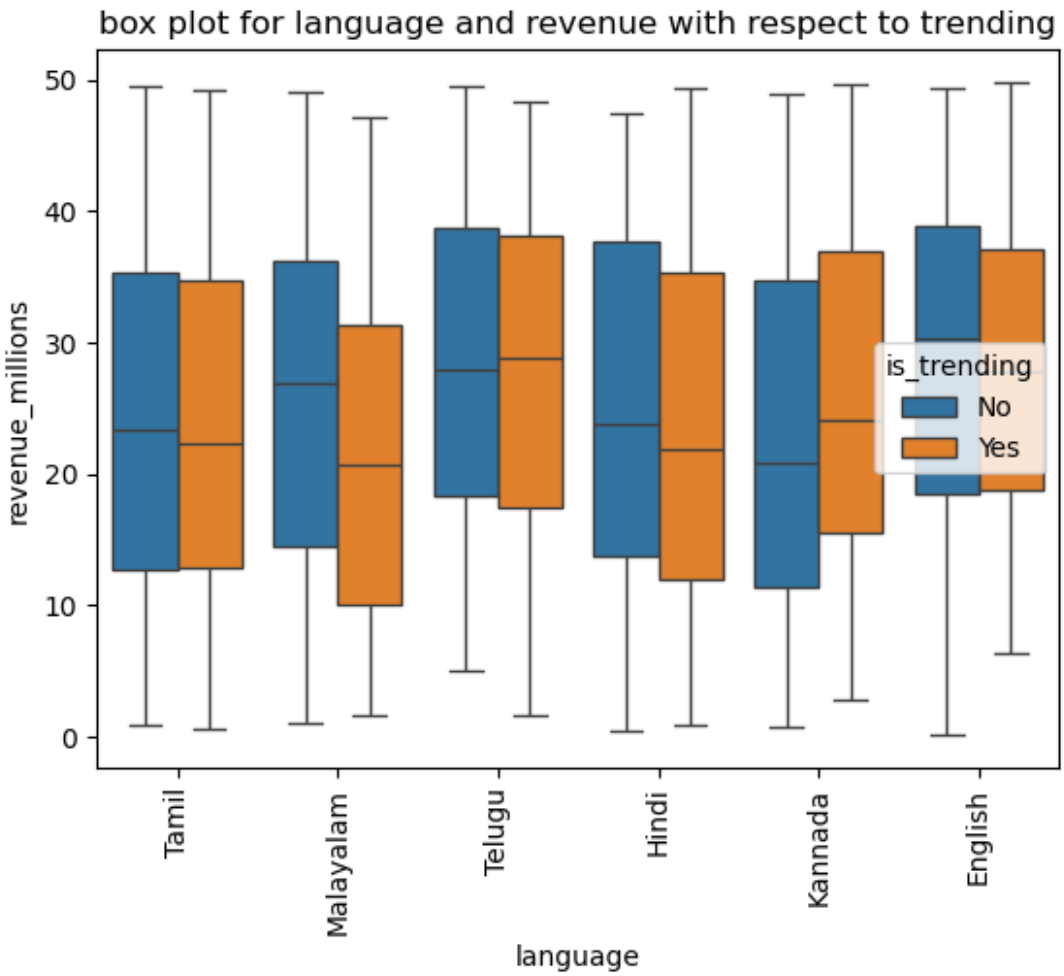
the box plot shows that the highest median genre as per view with respect to age rating like romance with age rating A as follow documentary A and lowest median is comedy in age rating U.

```
sns.boxplot(x="country",y="revenue_millions",hue="subscription_plan",d
ata=df)
plt.title("box plot for country and revenue with respect to
subscription plan")
plt.xticks(rotation =90)
plt.show()
```



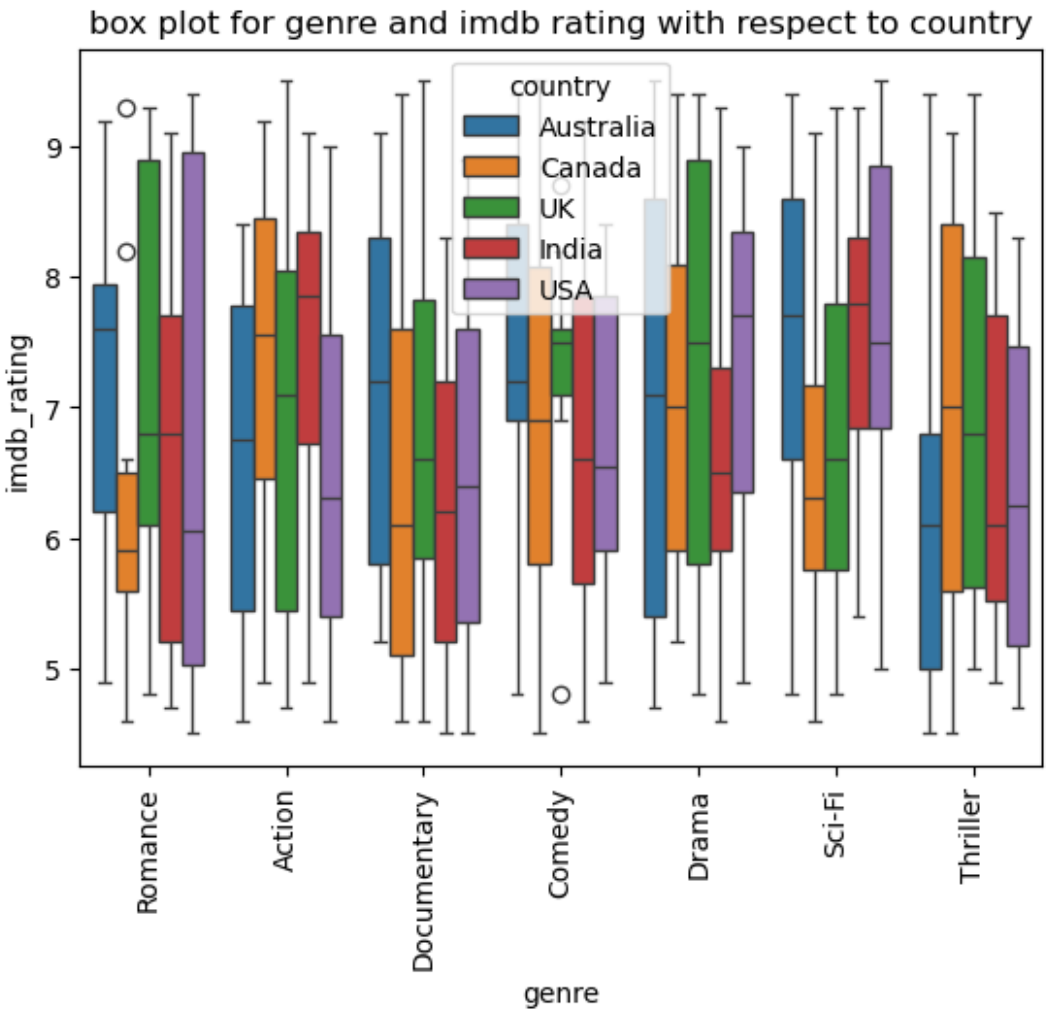
the box plot shows that the highest median of country with respect to subscription plan is Canada with subscription is premium and standard and lowest median is india with subscription standard as follow USA in premium.

```
sns.boxplot(x="language",y="revenue_millions",hue="is_trending",data=d
f)
plt.title("box plot for language and revenue with respect to
trending")
plt.xticks(rotation =90)
plt.show()
```



the box plot shows that the highest median of language as per trending is malayalam not trending and lowest median is kanada trending and not trending.

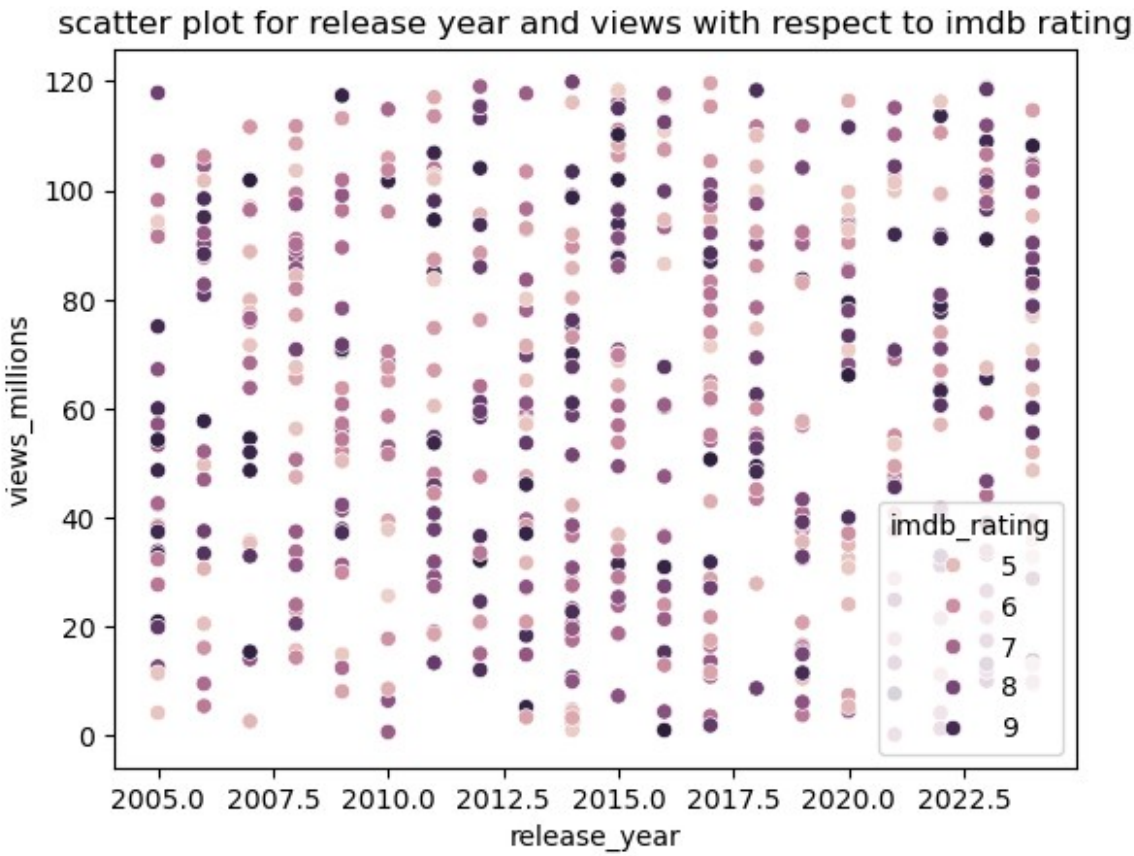
```
sns.boxplot(x="genre",y="imdb_rating",hue="country",data=df)
plt.title("box plot for genre and imdb rating with respect to
country")
plt.xticks(rotation =90)
plt.show()
```



the box plot shows that highest median of genre as per country is romance in Australia and india , action in Australia,canada,UK and india, Documentary in Australia, comedy in UK , drama in USA, sci-fi in india and thriller in Australia and lowest median romance in canada, action in USA , documentary in Uk and USA, comedy in Australia,USA, drama in india, sci-fi in canada and UK and thriller in india and USA.

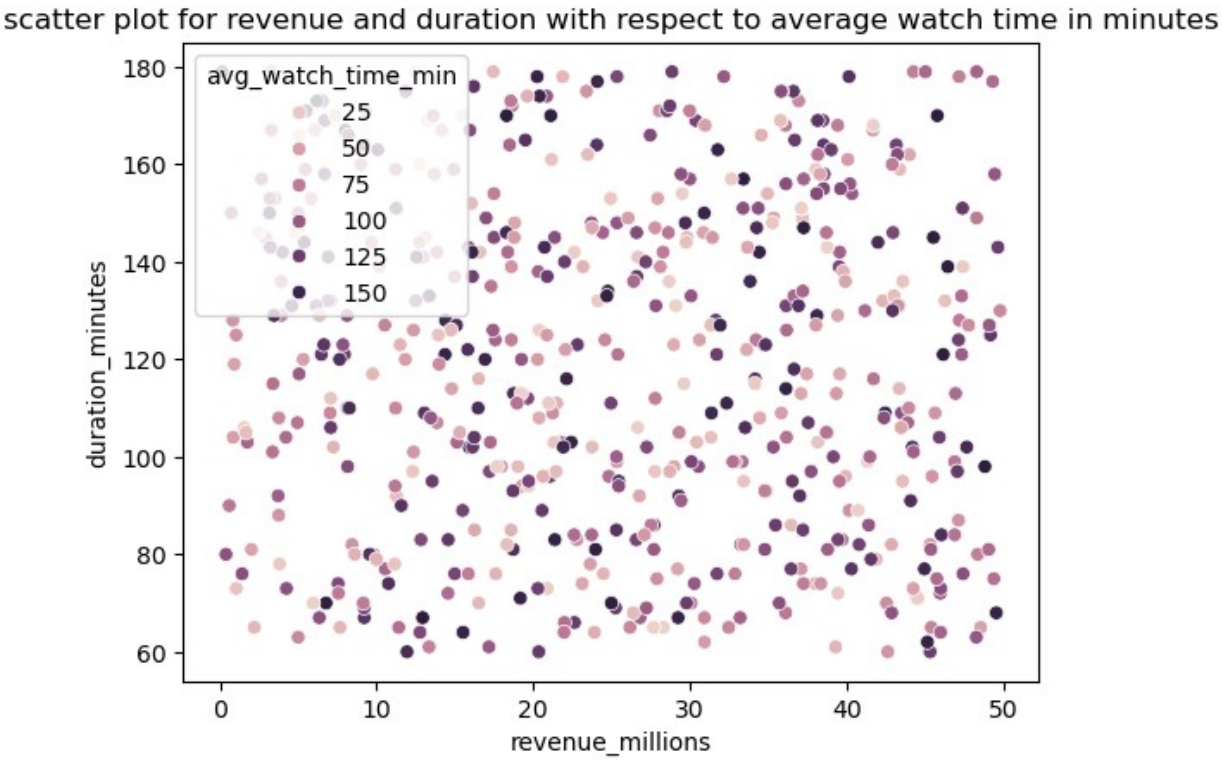
SCatter plot(bivariate)

```
sns.scatterplot(x="release_year",y="views_millions",hue="imdb_rating",
data=df)
plt.title("scatter plot for release year and views with respect to
imdb rating")
plt.show()
```



the scatter plot indicates that there is not ceratin changes among release year and views.there is certain movies at those year that get higher views.

```
sns.scatterplot(x="revenue_millions",y="duration_minutes",hue="avg_wat
ch_time_min",data=df)
plt.title("scatter plot for revenue and duration with respect to
average watch time in minutes")
plt.show()
```



the scatter plot shows that the there is no strong relationship between revenue and duration.it doesn't form any single line toh interpretated the data .